Security-Reliability Analysis of Backscatter Communications for Copyright-Protected Content Delivery in IoT Networks

Minh-Han H. Do[1], Hye-Young Kim[2], Thu-Ha Thi Pham[1], Tan N. Nguyen[3,*], Phuong T. Tran[3], Byung-Seo Kim[4], and Miroslav Voznak[5]

[1]Faculty of Electrical and Electronics Engineering, Ton Duc Thang University, Ho Chi Minh City, Vietnam.
[2]School of Games/Game Software, Hongik University, Sejong-si 30016, South Korea.
[3]Advanced Intelligent Technology Research Group, Faculty of Electrical and Electronics Engineering, Ton Duc Thang University, Ho Chi Minh City, Vietnam.
[4]Department of Software and Communications Engineering, Hongik University-Sejong Campus, 30016 South Korea.
[5]Faculty of Electrical Engineering and Computer Science, VSB–Technical University of Ostrava, 708 00 Ostrava, Czechia.
[*]nguyennhattan@tdtu.edu.vn

ABSTRACT

Secure backscatter communication (BackCom) is a promising paradigm for low-complexity, spectrum-efficient, and energy-efficient wireless networks. Owing to their passive transmission nature and physical-layer security potential, such systems offer intrinsic protection against unauthorized access, making them particularly suitable for securing digital content and upholding copyright in open, interference-prone wireless environments. However, their reliability and security can be severely affected by active full-duplex (FD) eavesdropping, in which an adversary simultaneously jams the legitimate receiver and attempts to decode the confidential transmission. This paper investigates a source-assisted BackCom system composed of a source, a backscatter device, a legitimate destination, and an active FD eavesdropper. The legitimate nodes are equipped with single antennas, whereas the eavesdropper employs separate transmit and receive antennas to jam the legitimate destination while intercepting confidential information under residual self-interference. By considering successive interference cancellation at both the legitimate destination and the eavesdropper, the signal-to-interference-plus-noise ratio expressions for the two decoding stages are derived. Based on these expressions, exact analytical formulations for the outage probability (OP) and intercept probability (IP) are obtained. The integral-form expressions are further transformed into closed-form representations in terms of the bivariate generalized Fox-H function. High signal-to-noise ratio (SNR) approximations are also derived, showing that both the OP and IP converge to non-zero asymptotic limits in the high-SNR regime. Monte Carlo simulations are conducted to validate the accuracy of the analytical results. Numerical results show that the backscatter coefficient has a non-monotonic effect on both reliability and interception performance, while the eavesdropper's jamming power increases the OP at the legitimate destination but reduces the IP due to residual self-interference. These results reveal an inherent reliability-security trade-off and provide practical design guidelines for enabling secure, reliable, and energy-efficient content delivery in next-generation wireless networks.

Keywords: Backscatter communication, physical-layer security, active eavesdropping, content security, copyright protection, digital content.

1 Introduction

1.1 Research Context

The integration of the Internet of Things (IoT) and fifth-generation (5G) wireless technologies has become a key foundation for next-generation connected systems[1]. Large-scale IoT deployments require massive connectivity, wide-area coverage, and stringent quality-of-service support[2]. Emerging smart-system applications further impose demanding requirements on network capacity, data rates, latency, and spectrum utilization. Meeting these requirements introduces additional challenges in intelligent network management[3] and the large-scale integration of IoT devices into 5G networks[4]. These requirements are particularly relevant to smart-city and industrial IoT applications, where large numbers of low-cost devices are expected to operate for extended periods with limited human intervention[5,6]. To support such deployments, long-range low-power wide-area network technologies, including LoRa, have been extensively investigated to provide energy-efficient connectivity

over wide coverage areas[7]. In parallel, reconfigurable intelligent surfaces (RISs) have emerged as a complementary approach for improving wireless coverage, propagation conditions, and energy efficiency through controllable radio environments[8]. Nevertheless, further reducing the energy consumption and hardware burden of resource-constrained IoT devices remains an important design objective, motivating communication architectures with increasingly passive and low-complexity end devices.

Against this background, backscatter communication (BackCom) has emerged as a promising paradigm for ultra-low-power and spectrum-efficient IoT connectivity. A comprehensive overview of its operating principles and potential applications is provided in[9]. Unlike conventional active radios, a backscatter device (BD) does not generate a dedicated radio-frequency (RF) carrier. Instead, it conveys information by modulating its antenna impedance and reflecting an incident RF waveform. Owing to this passive operation, BackCom can substantially reduce the active RF circuitry and power burden at the device side, making it particularly suitable for battery-limited and battery-free IoT applications, as discussed in[10]. Recent developments have further extended BackCom toward RIS-assisted architectures[11], while coding techniques tailored to backscatter transmission have been systematically investigated in[12]. Ambient backscatter communication (AmBC) further broadens this paradigm by reusing surrounding RF transmissions as carrier waveforms. Its integration with cellular networks has been investigated in[13], whereas Wi-Fi-based ambient backscatter was studied in[14]. The fundamental principles, implementation challenges, and emerging opportunities of AmBC are reviewed in[15]. By reducing the dependence on dedicated carrier generation and active transmission circuitry at the BD, AmBC is particularly attractive for dense and energy-constrained IoT deployments. Despite these advantages, the broadcast nature of wireless propagation exposes both direct and backscattered transmissions to potential unauthorized reception[16]. Moreover, the limited energy and computational resources of passive BDs can constrain the complexity of security processing implemented directly at the device. To address this issue, physical-layer security (PLS) has emerged as a complementary means of enhancing wireless confidentiality by exploiting channel fading, noise, interference, and differences between legitimate and eavesdropping links. The application of PLS to BackCom was investigated in[17], where the intrinsic characteristics of backscatter channels were exploited to improve secrecy performance. This framework was subsequently extended to multi-antenna BackCom systems in[18], with particular emphasis on artificial noise (AN) and resource optimization. More recent studies have explored PLS in wireless-powered AmBC networks[19] and vehicular communication systems assisted by ambient backscatter[20]. An AmBC-assisted mobile edge computing (MEC) framework with backscatter device selection was further studied in[21], demonstrating the potential of passive reflection for improving secure task offloading. From a different perspective, the use of RIS for enhancing PLS in vehicular networks was examined in[22]. A broader overview of PLS techniques for MEC enabled networks is provided in[23], highlighting the increasing relevance of physical layer protection for resource constrained wireless systems. Collectively, these studies show that PLS can complement conventional cryptographic and authentication mechanisms by providing an additional protection layer at the wireless interface.[1]

1.2 Literature Overview

The reliability and PLS of low-power wireless and backscatter-assisted systems have been investigated under increasingly sophisticated communication and threat models. Early studies mainly focused on passive eavesdropping or reliability analysis without an active adversarial transmission. For instance, the outage and intercept behaviors of energy-constrained wireless networks were analyzed in[24] for half-duplex energy-harvesting relaying, where the final performance expressions retain integral terms and the eavesdropper acts only as a passive receiver. In the BackCom context, the device-to-device partial non-orthogonal multiple access (NOMA) framework in[25] investigated outage performance and spectral efficiency but did not incorporate an eavesdropper. More recent studies have extended reliability and security analysis to more involved backscatter architectures. A cooperative backscatter-NOMA network with multiple tags and multiple colluding or non-colluding eavesdroppers was studied in[26], where closed-form outage probability (OP) and intercept probability (IP) expressions together with high signal-to-noise ratio (SNR) approximations and diversity analysis were developed. In[27], randomly distributed passive eavesdroppers were incorporated into an AmBC-NOMA network using stochastic geometry, while AN generated by the base station (BS) and an eavesdropper-exclusion region were introduced to improve PLS. The corresponding OP and IP behaviors and their high-SNR characteristics were then analyzed. Cognitive AmBC has also been investigated from the security-reliability perspective. The work in[28] characterized the OP and IP of the source and backscatter transmissions, whereas[29] further introduced transmit-antenna selection and derived closed-form OP/IP expressions and asymptotic results under Nakagami-m fading. Although these studies provide increasingly comprehensive probabilistic characterizations of BackCom reliability and security, their eavesdroppers remain passive. Consequently, the adversary does not intentionally degrade legitimate reception, and its own interception performance is not coupled to a simultaneously transmitted attack signal.

A second line of research has enhanced BackCom security through cooperative transmission and friendly jamming. Wireless-powered cooperative AmBC was investigated in[19], where a backscatter device cooperates with a decode-and-forward relay

[1] In this work, copyright-protected content delivery refers specifically to protecting copyrighted or access-restricted digital content against unauthorized over-the-air recovery during wireless transmission. PLS is therefore considered as a complementary confidentiality mechanism at the wireless interface rather than as a replacement for higher-layer copyright enforcement, authentication, or cryptographic access control.

and secrecy outage probability (SOP) and secrecy energy efficiency were used to quantify the resulting security performance. A particularly relevant study is[30], which considered an energy-harvesting-enabled AmBC architecture for the protection of digital content and copyrights. In that framework, a multi-antenna power beacon supplies energy and simultaneously acts as a friendly jammer, while exact closed-form OP and IP expressions were derived separately for the source and BD transmissions and complemented by high-SNR analysis. The destination and eavesdropper successively decode the source and backscatter components, which makes this work closely related to the transmission structure considered here. The fundamental threat model, however, is different because the eavesdropper in[30] remains passive and the jamming waveform originates from a legitimate power beacon. Therefore, the node producing interference and the node attempting interception are distinct. An unmanned aerial vehicle (UAV)-assisted AmBC system with a multi-antenna cooperative jammer was subsequently investigated in[31]. The SOPs of the air-ground and backscatter links were evaluated through a two-layer Gaussian-Chebyshev quadrature approach, followed by high-SNR asymptotic analysis. In these defensive-jamming architectures, the jamming signal is deliberately introduced by a legitimate node to suppress interception. Hence, they do not produce the adversarial coupling considered in this work, where the same eavesdropper transmission that degrades the legitimate destination also influences the eavesdropper's own reception through residual self-interference (RSI).

Active and proactive eavesdropping has also been investigated, although the associated architectures and research objectives differ substantially from the present source-assisted BackCom analysis. Outside the BackCom context,[32] considered a full-duplex (FD) space-air-ground integrated network (SAGIN) with passive and active eavesdroppers (AEs) together with a friendly jammer. OP, IP, and transmission SOP were analyzed to quantify the effect of the active attack and the defensive jammer. This study establishes the importance of probabilistic reliability and security analysis under active eavesdropping. Nevertheless, its transmission architecture is based on conventional aerial relaying and does not involve an information-bearing backscatter link, cascaded source-BD channels, or successive recovery of source and backscatter components. Within the BackCom literature,[33] investigated a full-duplex symbiotic-radio system subject to a proactive eavesdropper that simultaneously intercepts the legitimate transmission and emits an attack signal. Importantly, both direct attack propagation and attack components coupled through the backscatter link are incorporated in that model. AN generation and forward-noise suppression are then jointly designed, and phase control and power allocation are optimized to maximize security throughput. In[34], an RIS-assisted backscatter architecture was proposed to exploit the AE's jamming waveform for enhancing legitimate reception. The resulting problem maximizes the intended receiver signal-to-interference-plus-noise ratio (SINR) through alternating optimization under source-power, RIS-reflection, and eavesdropper-SINR constraints. Proactive eavesdropping was also investigated for passive radio-frequency identification (RFID) communication in[35], where the eavesdropper broadcasts a continuous-wave signal to simultaneously strengthen wiretapping and interfere with the legitimate reader. A wiretap-channel-aware AN strategy, transmit-power allocation, and a hierarchical security game were developed to maximize the secrecy rate. These studies demonstrate that adversary-originated interference has already been recognized as an important threat in backscatter-related systems. Their emphasis, however, is predominantly on attack mitigation, beamforming, AN design, reflection optimization, or secrecy-rate maximization rather than on deriving a joint probabilistic reliability and interception framework for the information-bearing source and backscatter components.

Taken together, the existing literature leaves a more specific analytical gap than the individual consideration of BackCom, OP/IP analysis, or active eavesdropping alone. The reviewed passive-eavesdropper BackCom studies provide detailed reliability-security characterizations but do not capture adversary-originated jamming. Friendly-jamming approaches introduce additional interference as a legitimate defensive mechanism, so the jammer and eavesdropper remain separate entities. Existing active-eavesdropping studies establish the relevance of adversarial transmission and, in some cases, explicitly account for attack signals propagated through backscatter paths, but they mainly address security enhancement and resource optimization rather than the coupled OP and IP behavior considered here. In contrast, the present work considers a source-assisted BackCom architecture in which an active FD eavesdropper simultaneously transmits a jamming waveform toward the legitimate destination and attempts to recover two complementary information components carried by the source and the BD. The eavesdropper is therefore both the origin of the malicious interference and an information receiver affected by residual self-interference. Moreover, the destination and the eavesdropper both employ two-stage SIC, and complete legitimate or unauthorized content recovery requires successful decoding of both the source and backscatter components. This structure leads to joint decoding events involving direct fading, cascaded backscatter fading, adversarial jamming, and residual self-interference. Importantly, the two SIC stages are statistically coupled through common channel and interference terms, so the resulting OP and IP cannot be obtained by simply combining independent single-link decoding probabilities. The analysis must therefore characterize the joint feasible decoding regions of the source and backscatter components while accounting for the shared cascaded fading and adversarial-interference variables. Based on these coupled events, we derive exact OP and IP formulations, obtain closed-form representations involving the bivariate generalized Fox-H function, and develop high-SNR expressions that characterize the corresponding reliability floor and interception ceiling. Table 1 summarizes these distinctions from representative prior studies in terms of the threat model, jamming origin, performance metric, and analytical or design methodology.

Table 1. Comparison between the proposed framework and representative studies on the reliability and PLS of backscatter-assisted and active-eavesdropping wireless systems. EJ: eavesdropper-generated jamming.

Work	System	BC	AE	EJ	2-SIC	OP	IP	RSI	Techniques / Contributions
[26]	Cooperative backscatter-NOMA	✓	–	–	–	✓	✓	–	Closed-form OP/IP analysis for multiple tags and passive colluding/non-colluding eavesdroppers, with high-SNR and diversity characterization
[27]	AmBC-NOMA	✓	–	–	–	✓	✓	–	Stochastic-geometry OP/IP analysis with randomly located passive eavesdroppers, BS-generated AN, and an eavesdropper-exclusion region
[30]	Energy-harvesting AmBC	✓	–	–	✓	✓	✓	–	Exact source/BD OP and IP expressions with cascaded fading, SIC, friendly jamming from a legitimate power beacon, and high-SNR characterization
[31]	UAV-assisted AmBC	✓	–	–	–	–	–	–	SOP analysis using two-layer Gaussian–Chebyshev quadrature and high-SNR approximations with a cooperative multi-antenna jammer
[32]	FD SAGIN	–	✓	–	–	✓	✓	–	Reliability and secrecy analysis with active/passive eavesdroppers, UAV-relay selection, and a separate friendly jammer
[33]	FD symbiotic radio	✓	✓	✓	–	–	–	–	Proactive-eavesdropping model with attack signaling, AN-assisted defense, pseudo-decoding, and joint phase/power optimization
[34]	RIS-assisted BackCom	✓	✓	✓	–	–	–	–	Alternating optimization of source transmission and RIS reflection by exploiting the AE's jamming signal
[35]	RFID / passive-tag system	✓	✓	✓	–	–	–	–	Wiretap-channel-aware AN design, transmit-power allocation, and hierarchical security game against a proactive eavesdropper
Ours	**Source-assisted BackCom**	✓	✓	✓	✓	✓	✓	✓	**Joint reliability–security characterization with active FD eavesdropping, adversarial jamming, two-stage SIC, cascaded BackCom, RSI, exact integral/closed-form analysis, and high-SNR characterization**

1.3 Novelty and Contributions

Motivated by the above gap, this work investigates the reliability and security of source-assisted BackCom for protected wireless content delivery in the presence of an active FD eavesdropper. Unlike the closely related passive-eavesdropper BackCom frameworks and the active-eavesdropper systems summarized in Table 1, the considered adversary simultaneously degrades legitimate reception through jamming and attempts to recover the source and backscatter components while suffering from residual self-interference. The contribution is therefore centered on the coupled reliability and interception behavior created by the interaction of active jamming, cascaded backscatter propagation, and two-stage decoding. The main contributions are summarized as follows:

- We develop a source-assisted BackCom model for protected content delivery under active FD eavesdropping. At the service abstraction considered in this work, each protected content unit is represented by two complementary information components mapped to the source and backscatter transmissions. Complete legitimate recovery and complete unauthorized recovery therefore require successful decoding of both components. This assumption establishes a direct and explicit connection between the content-delivery model and the joint OP and IP definitions, while PLS is regarded only as a wireless confidentiality mechanism complementary to higher-layer security and copyright-enforcement techniques.

- We derive exact analytical expressions for the joint OP and IP by accounting for the direct links, cascaded source-BD-receiver channels, adversarial jamming, and residual self-interference at the active FD eavesdropper. The resulting integrals are further represented in closed form using the bivariate generalized Fox-H function. High-SNR expressions are then developed for the regime in which the normalized source transmit power increases while the adversarial jamming level remains fixed. These expressions reveal the resulting reliability floor and interception ceiling and clarify why increasing source power alone cannot remove the interference-limited behavior.

- We provide a numerical evaluation of the reliability-security trade-off with respect to normalized source transmit power, normalized eavesdropper jamming power, reflection coefficient, and independent source and backscatter target rates. The analysis is complemented by direct-transmission and passive-eavesdropping benchmarks, imperfect-SIC simulations, and an imperfect channel state information (CSI) sensitivity study. The backscatter coefficient is further examined through reliability-oriented and interception-critical operating points. These results quantify the conditions under which backscatter transmission makes complete interception more difficult, while also revealing the associated reliability cost and the sensitivity of the operating point to practical receiver imperfections.

The exact analytical framework adopts perfect SIC and a tractable statistical model for residual self-interference in order to obtain interpretable OP and IP expressions. These assumptions should therefore be regarded as analytical reference conditions rather than idealized descriptions of practical FD receivers. Receiver imperfections associated with channel-estimation errors

and residual cancellation are examined numerically, whereas a unified closed-form treatment incorporating imperfect SIC and hardware-dependent self-interference remains an important extension of the present framework.

1.4 Organization

The remainder of this paper is organized as follows. Section 2 presents the considered system model and derives the corresponding SINR expressions. Section 3 analyzes the OP and its high-SNR asymptotic behavior. Section 4 characterizes the IP and derives the corresponding high-SNR asymptotic expression. Section 5 provides numerical and Monte Carlo simulation results to validate the analysis and discuss key design insights. Finally, Section 6 concludes the paper.

2 System Model

As illustrated in Fig. 1, we consider a secure source-assisted BC network consisting of one source node S, one BD, one legitimate destination D, and one active FD eavesdropper E. The legitimate nodes, including S, the BD, and D, are each equipped with a single antenna. In contrast, the eavesdropper E is equipped with two antennas, or equivalently two RF chains, to support FD operation: one receiving antenna is used for eavesdropping, while one transmitting antenna is used for emitting a jamming signal toward the legitimate destination. Here, "active" indicates that E not only intercepts the ongoing transmission but also deliberately interferes with the legitimate link through simultaneous jamming, consistent with the proactive-eavesdropping operation considered in related full-duplex symbiotic-radio systems[33]. The source continuously emits an RF signal to support both the direct transmission to D and the passive backscatter transmission from the BD. The BD is a passive single-antenna device and conveys its own information by modulating its antenna impedance and reflecting the incident RF signal from S.

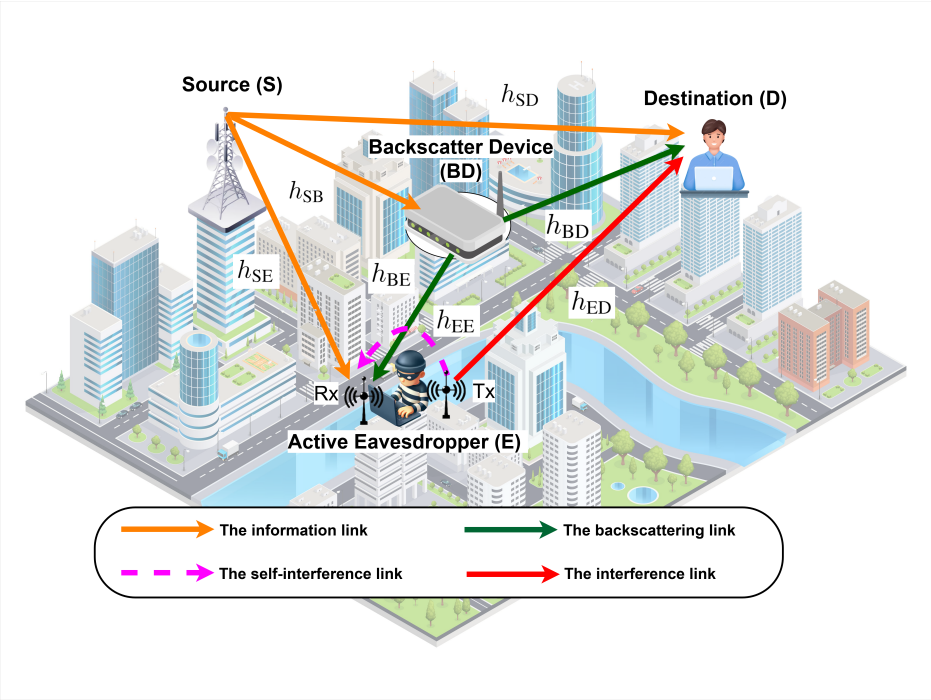


Figure 1. System model of the considered secure source-assisted BC network with an active FD eavesdropper.

The received signals at D and E are respectively given by

$$y_{\mathrm{D}} = \sqrt{P_{\mathrm{S}}} h_{\mathrm{SD}} x_{\mathrm{S}} + \sqrt{\beta P_{\mathrm{S}}} h_{\mathrm{SB}} h_{\mathrm{BD}} x_{\mathrm{S}} x_{\mathrm{B}} + \sqrt{P_{\mathrm{E}}} h_{\mathrm{ED}} x_{\mathrm{E}} + n_{\mathrm{D}}, \tag{1}$$

and

$$y_{\mathrm{E}} = \sqrt{P_{\mathrm{S}}} h_{\mathrm{SE}} x_{\mathrm{S}} + \sqrt{\beta P_{\mathrm{S}}} h_{\mathrm{SB}} h_{\mathrm{BE}} x_{\mathrm{S}} x_{\mathrm{B}} + \sqrt{P_{\mathrm{E}}} h_{\mathrm{EE}} x_{\mathrm{E}} + n_{\mathrm{E}}. \tag{2}$$

Here, x_{S}, x_{B}, and x_{E} denote the source information symbol, the backscatter information symbol, and the jamming symbol generated by E, respectively. The source, backscatter, and jamming symbols are assumed to be mutually independent and zero mean, with $\mathbb{E}\{|x_{\mathrm{S}}|^2\} = \mathbb{E}\{|x_{\mathrm{B}}|^2\} = \mathbb{E}\{|x_{\mathrm{E}}|^2\} = 1$. Constant-envelope phase shift keying (PSK) signalling is adopted for

the information-bearing symbols, such that $|x_S|^2 = |x_B|^2 = 1$. For example, x_S and x_B may be drawn from independently encoded binary phase shift keying (BPSK) or quadrature phase shift keying (QPSK) constellations, while their target spectral efficiencies R_S and R_B are determined by the corresponding channel codes. No particular constellation is required for x_E; only its independence from the information-bearing symbols and its unit average power are used in the SINR analysis. The source and BD transmissions are assumed to be symbol-synchronous. In particular, the BD maintains a constant reflection state during each source-symbol interval and changes its state only at a symbol boundary. Perfect timing and carrier synchronisation are assumed at both receivers.

The quantities P_S and P_E denote the physical transmit powers of the source and the AE, respectively, whereas $\beta \in (0,1)$ denotes the effective backscatter coefficient. The noise components at receivers D and E are modeled as $n_D, n_E \sim \mathscr{CN}(0, N_0)$, following a circularly symmetric complex Gaussian (CSCG) distribution, where N_0 denotes the noise variance of the additive white Gaussian noise (AWGN). For subsequent analysis, we define $\rho_S \triangleq P_S/N_0$ and $\rho_E \triangleq P_E/N_0$, referred to as the normalized source transmit power and the normalized eavesdropper jamming power, respectively. In the considered protected-content delivery model, both x_S and x_B are information-bearing and convey complementary components of the same protected content unit. Specifically, x_S carries the source information component through the direct transmission, whereas x_B carries the complementary backscatter information component by modulating the reflection of the incident source waveform, resulting in the multiplicative term $x_S x_B$. Under the adopted service abstraction, neither component alone constitutes complete recovery of the protected content. Accordingly, complete legitimate recovery at D and complete unauthorized recovery at E require successful decoding of both components. This application-level mapping defines the joint decoding events employed in the subsequent OP and IP analyses.

Next, the channel coefficients h_{SB}, h_{SD}, h_{SE}, h_{BD}, h_{BE}, h_{ED}, and h_{EE} correspond to the S $\rightarrow$BD, S $\rightarrow$D, S $\rightarrow$E, BD$\rightarrow$ D, BD$\rightarrow$ E, E $\rightarrow$D, and residual self-interference links, respectively. More specifically, h_{SE} and h_{BE} denote the channels from S and the BD to the receiving antenna of E, respectively, whereas h_{ED} denotes the jamming link from the transmitting antenna of E to D. The coefficient h_{EE} models the residual loopback self-interference channel from the transmitting antenna to the receiving antenna of the FD eavesdropper after self-interference suppression. The noise generated at the passive BD is neglected in the reflected signal, as commonly assumed in BC systems. The external propagation channels are assumed to undergo independent, but not necessarily identically distributed, Rayleigh fading. The effective post-cancellation RSI coefficient h_{EE} is also modelled as an independent zero-mean complex Gaussian random variable for analytical tractability. Accordingly, $\gamma_i \triangleq |h_i|^2 \sim \mathrm{Exp}(\lambda_i), i \in \{SB, SD, SE, BD, BE, ED, EE\}$, where λ_i is the corresponding rate parameter. For notational compactness, the cascaded backscatter channel gains are defined as $\gamma_{SBD} \triangleq \gamma_{SB}\gamma_{BD}$ and $\gamma_{SBE} \triangleq \gamma_{SB}\gamma_{BE}$. Correspondingly, we define the composite channel parameters as $\lambda_{SBD} \triangleq \lambda_{SB}\lambda_{BD}$ and $\lambda_{SBE} \triangleq \lambda_{SB}\lambda_{BE}$.

The received-signal model in (1) and (2) retains the source-excited backscatter components and neglects jammer-originated reflections. The latter would introduce $\sqrt{\beta P_E} h_{EB} h_{BD} x_E x_B$ at D and $\sqrt{\beta P_E} h_{EB} h_{BE} x_E x_B$ at E, where h_{EB} denotes the channel from the transmitting antenna of E to the BD. We consider a source-dominated illumination regime characterized by

$$P_E \mathbb{E}\left[|h_{EB}|^2\right] \ll P_S \mathbb{E}\left[|h_{SB}|^2\right]. \tag{3}$$

The expectations are taken over channel fading, so (3) specifies dominance in terms of the average powers incident on the BD. The omission of the reflected jamming terms additionally assumes $\beta P_E \mathbb{E}\left[|h_{EB}h_{BD}|^2\right] \ll P_E \mathbb{E}\left[|h_{ED}|^2\right] + N_0$, $\beta P_E \mathbb{E}\left[|h_{EB}h_{BE}|^2\right] \ll P_E \mathbb{E}\left[|h_{EE}|^2\right] + N_0$. These inequalities compare the average reflected jamming powers with the retained interference-plus-noise powers at D and E, respectively. The comparison at E uses the residual self-interference after suppression. Such a regime may arise from strong attenuation or directional isolation on the E $\rightarrow$ BD link and does not require $P_E < P_S$. A related omission of jammer-originated backscatter under weak illumination is adopted in[36]. When reflected jamming is appreciable relative to the retained interference and noise, the additional components must be included.

A two-stage SIC procedure is employed at both D and E. The source symbol x_S is decoded first because the direct source component is assumed to dominate the backscattered component in average received power. This assumption is reasonable because the backscattered signal experiences both the reflection loss $\beta < 1$ and the cascaded propagation loss over the S$\rightarrow$BD$\rightarrow k$ path, where $k \in \{D, E\}$. During the first decoding stage, the backscatter symbol x_B is unknown. Hence, the multiplicative term $x_S x_B$ cannot yet be reconstructed or cancelled and is treated as an unknown co-channel interference component. Once x_S has been decoded successfully, its direct contribution is removed through SIC. The decoded source symbol then provides the carrier reference required to detect x_B from the remaining backscattered component. After successful decoding of the source symbol, perfect SIC is assumed[2], such that the source component is completely cancelled before decoding the backscatter symbol. Consequently, during the second decoding stage, only the residual jamming signal and the thermal noise remain as interference. Accordingly, based on equation (1), the received SINR for decoding the source symbol at

[2]The perfect SIC assumption is adopted as an ideal decoding benchmark to isolate the effects of active FD eavesdropping and cascaded backscatter fading.

D is expressed as

$$\mathrm{SINR}_{\mathrm{D},1} = \frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SD}}}{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SB}}\gamma_{\mathrm{BD}} + \rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1} = \frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SD}}}{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBD}} + \rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1}, \tag{4}$$

where the desired signal is the direct transmission from S, while the backscatter signal, the artificial jamming signal from E, and the receiver noise constitute the interference terms. After applying perfect SIC, the SINR for decoding the backscatter symbol becomes

$$\mathrm{SINR}_{\mathrm{D},2} = \frac{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBD}}}{\rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1}. \tag{5}$$

Following the same SIC procedure as that at D, the eavesdropper E first decodes the source signal by treating the backscatter signal and its residual self-interference as interference. After successful source decoding and cancellation, the backscatter signal is subsequently detected, consistent with the successive source-and-backscatter decoding structure adopted in related secured BackCom systems[30].

$$\mathrm{SINR}_{\mathrm{E},1} = \frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SE}}}{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SB}}\gamma_{\mathrm{BE}} + \rho_{\mathrm{E}}\gamma_{\mathrm{EE}} + 1} = \frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SE}}}{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBE}} + \rho_{\mathrm{E}}\gamma_{\mathrm{EE}} + 1}, \tag{6}$$

$$\mathrm{SINR}_{\mathrm{E},2} = \frac{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBE}}}{\rho_{\mathrm{E}}\gamma_{\mathrm{EE}} + 1}. \tag{7}$$

It is observed that the normalized eavesdropper jamming power ρ_{E} has two opposite effects. A larger ρ_{E} increases the interference observed at the legitimate destination through the E $\rightarrow$ D link, thereby degrading its decoding performance. At the same time, it increases the residual self-interference at the receiving antenna of the full-duplex eavesdropper and consequently impairs its own interception capability.

3 Reliability Analysis

3.1 Outage Probability Analysis

In this section, we analyze the reliability performance at the legitimate destination D. Let R_{S} and R_{B}, measured in bit/s/Hz, denote the target data rates associated with the source signal and the backscatter signal, respectively. Since the source and backscatter signals are conveyed through different transmission mechanisms and may be associated with different rate requirements, separate decoding thresholds are assigned to the two SIC stages. Specifically, $\gamma_{\mathrm{th}} = 2^{R_{\mathrm{S}}} - 1, \widetilde{\gamma}_{\mathrm{th}} = 2^{R_{\mathrm{B}}} - 1$, where γ_{th} is the decoding threshold for the source signal and $\widetilde{\gamma}_{\mathrm{th}}$ is that for the backscatter signal. This formulation permits different rate requirements for the direct and backscatter transmissions and includes the equal-rate case as the special case $R_{\mathrm{S}} = R_{\mathrm{B}}$. Accordingly, the OP is formulated as

$$\begin{aligned}
\mathrm{OP} &= 1 - \Pr\left(\mathrm{SINR}_{\mathrm{D},1} \geq \gamma_{\mathrm{th}}, \mathrm{SINR}_{\mathrm{D},2} \geq \widetilde{\gamma}_{\mathrm{th}}\right) \\
&= 1 - \Pr\left(\frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SD}}}{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBD}} + \rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1} \geq \gamma_{\mathrm{th}}, \frac{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBD}}}{\rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1} \geq \widetilde{\gamma}_{\mathrm{th}}\right) \\
&= 1 - \int_0^{\infty} \underbrace{\Pr\left(\frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SD}}}{\beta\rho_{\mathrm{S}}x + \rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1} \geq \gamma_{\mathrm{th}}, \frac{\beta\rho_{\mathrm{S}}x}{\rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1} \geq \widetilde{\gamma}_{\mathrm{th}}\right)}_{\Theta} f_{\gamma_{\mathrm{SBD}}}(x)\,dx.
\end{aligned} \tag{8}$$

From (8), the conditional successful decoding probability can be expressed as

$$\Theta = \Pr\left(\frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SD}}}{\beta\rho_{\mathrm{S}}x + \rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1} \geq \gamma_{\mathrm{th}}, \frac{\beta\rho_{\mathrm{S}}x}{\rho_{\mathrm{E}}\gamma_{\mathrm{ED}} + 1} \geq \widetilde{\gamma}_{\mathrm{th}}\right). \tag{9}$$

The second-stage decoding condition yields $\frac{\beta\rho_{\mathrm{S}}x}{\rho_{\mathrm{E}}\gamma_{\mathrm{ED}}+1} \geq \widetilde{\gamma}_{\mathrm{th}} \Longleftrightarrow \gamma_{\mathrm{ED}} \leq \xi$, where $\xi = \frac{\beta\rho_{\mathrm{S}}x - \widetilde{\gamma}_{\mathrm{th}}}{\widetilde{\gamma}_{\mathrm{th}}\rho_{\mathrm{E}}}$. Since γ_{ED} is non-negative, a feasible decoding region exists only when $\xi \geq 0$, which is equivalent to $x \geq \tau$, where $\tau = \frac{\widetilde{\gamma}_{\mathrm{th}}}{\beta\rho_{\mathrm{S}}}$. Accordingly, conditioned on $\gamma_{\mathrm{SBD}} = x$ with $x \geq \tau$, the successful decoding probability Θ can be obtained by averaging over the PDF of γ_{ED} as

$$\Theta = \int_0^{\xi} \Pr\left(\gamma_{SD} \geq \frac{\gamma_{th}(\beta\rho_S x + \rho_E y + 1)}{\rho_S}\right) f_{\gamma_{ED}}(y)\, dy$$
$$= \int_0^{\xi} \left[1 - F_{\gamma_{SD}}\left(\frac{\gamma_{th}(\beta\rho_S x + \rho_E y + 1)}{\rho_S}\right)\right] f_{\gamma_{ED}}(y)\, dy. \tag{10}$$

Using the exponential distributions[37] of γ_{SD} and γ_{ED}, (10) becomes

$$\Theta = \exp\left[-\frac{\lambda_{SD}\gamma_{th}(\beta\rho_S x + 1)}{\rho_S}\right] \times \int_0^{\xi} \lambda_{ED} \exp\left[-\left(\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}\right) y\right] dy. \tag{11}$$

Evaluating the integral in (11), we obtain

$$\Theta = \exp\left[-\frac{\lambda_{SD}\gamma_{th}(\beta\rho_S x + 1)}{\rho_S}\right] \times \frac{\lambda_{ED}}{\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}} \times \left[1 - \exp\left(-\left[\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}\right]\xi\right)\right]. \tag{12}$$

Substituting (12) into (8), the OP is rewritten as

$$\mathrm{OP} = 1 - \frac{\lambda_{ED}}{\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}} \int_{\tau}^{\infty} \exp\left[-\frac{\lambda_{SD}\gamma_{th}(\beta\rho_S x + 1)}{\rho_S}\right] \times \left[1 - \exp\left(-\left[\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}\right]\xi\right)\right] f_{\gamma_{SBD}}(x)\, dx. \tag{13}$$

Since γ_{SB} and γ_{BD} are independent exponential random variables, the probability density function of γ_{SBD} is given by[38]

$$f_{\gamma_{SBD}}(x) = 2\lambda_{SBD} K_0\left(2\sqrt{\lambda_{SBD} x}\right), \qquad x \geq 0. \tag{14}$$

where $K_0(\cdot)$ denotes the modified Bessel function of the second kind of order zero. Substituting (14) into (13), we obtain

$$\mathrm{OP} = 1 - \frac{\lambda_{ED}}{\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}} \int_{\tau}^{\infty} \left[1 - \exp\left(-\left[\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}\right]\xi\right)\right]$$
$$\times \exp\left[-\frac{\lambda_{SD}\gamma_{th}(\beta\rho_S x + 1)}{\rho_S}\right] \times 2\lambda_{SBD} K_0\left(2\sqrt{\lambda_{SBD} x}\right) dx. \tag{15}$$

By expanding the term inside the square brackets, the exact OP is expressed as

$$\mathrm{OP} = 1 - \frac{\lambda_{ED}}{\lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}} \left[\exp\left(-\frac{\lambda_{SD}\gamma_{th}}{\rho_S}\right) \int_{\tau}^{\infty} \exp\left(-\lambda_{SD}\beta\gamma_{th} x\right) 2\lambda_{SBD} K_0\left(2\sqrt{\lambda_{SBD} x}\right) dx\right.$$
$$\left. - \exp\left(\frac{\lambda_{ED}}{\rho_E}\right) \int_{\tau}^{\infty} \exp\left(-\left[\lambda_{SD}\beta\gamma_{th}\left(1 + \frac{1}{\widetilde{\gamma}_{th}}\right) + \frac{\lambda_{ED}\beta\rho_S}{\widetilde{\gamma}_{th}\rho_E}\right] x\right) 2\lambda_{SBD} K_0\left(2\sqrt{\lambda_{SBD} x}\right) dx\right]. \tag{16}$$

For analytical tractability, we first rewrite (16) in a more compact form. Define $\mathscr{A} \triangleq \lambda_{ED} + \lambda_{SD}\gamma_{th}\frac{\rho_E}{\rho_S}$, $c_1 \triangleq \lambda_{SD}\beta\gamma_{th}$, and $c_2 \triangleq \lambda_{SD}\beta\gamma_{th}\left(1 + \frac{1}{\widetilde{\gamma}_{th}}\right) + \frac{\lambda_{ED}\beta\rho_S}{\widetilde{\gamma}_{th}\rho_E}$.

Then, (16) can be rewritten as

$$\mathrm{OP} = 1 - \frac{2\lambda_{SBD}\lambda_{ED}}{\mathscr{A}} \Bigg(\underbrace{\exp\left(-\frac{\lambda_{SD}\gamma_{th}}{\rho_S}\right) \int_{\tau}^{\infty} e^{-c_1 x} K_0\left(2\sqrt{\lambda_{SBD} x}\right) dx}_{\mathscr{Y}_1} - \underbrace{\exp\left(\frac{\lambda_{ED}}{\rho_E}\right) \int_{\tau}^{\infty} e^{-c_2 x} K_0\left(2\sqrt{\lambda_{SBD} x}\right) dx}_{\mathscr{Y}_2} \Bigg). \tag{17}$$

To evaluate $\mathscr{Y}_1$ and $\mathscr{Y}_2$, we apply the change of variable $x = \tau t$, which yields $dx = \tau dt$. Accordingly, the lower integration limit $x = \tau$ is mapped to $t = 1$. Hence, for $i \in \{1,2\}$,

$$\mathscr{Y}_i = \tau e^{\omega_i} \int_1^{\infty} e^{-c_i \tau t} K_0\left(2\sqrt{\lambda_{\mathrm{SBD}} \tau t}\right) dt, \qquad i \in \{1,2\}, \tag{18}$$

where $\omega_1 = -\frac{\lambda_{\mathrm{SD}} \gamma_{\mathrm{th}}}{\rho_{\mathrm{S}}}, \omega_2 = \frac{\lambda_{\mathrm{ED}}}{\rho_{\mathrm{E}}}$.

Using the Heaviside step function[39] $H(t-1)$, the integral over $[1,+\infty)$ can be equivalently expressed over $[0,+\infty)$ as

$$\mathscr{Y}_i = \tau e^{\omega_i} \int_0^{\infty} e^{-c_i \tau t} K_0\left(2\sqrt{\lambda_{\mathrm{SBD}} \tau t}\right) H(t-1)\, dt. \tag{19}$$

The exponential function[37], the modified Bessel function[40], and the Heaviside step function[39] can be represented in terms of Meijer-G functions[41] as

$$\exp(-c_i \tau t) = G_{0,1}^{1,0}\left(c_i \tau t \,\middle|\, \begin{matrix} - \\ 0 \end{matrix}\right), \tag{20}$$

$$K_0\left(2\sqrt{\lambda_{\mathrm{SBD}} \tau t}\right) = \frac{1}{2} G_{0,2}^{2,0}\left(\lambda_{\mathrm{SBD}} \tau t \,\middle|\, \begin{matrix} - \\ 0,0 \end{matrix}\right), \tag{21}$$

and

$$H(t-1) = G_{1,1}^{0,1}\left(t \,\middle|\, \begin{matrix} 1 \\ 0 \end{matrix}\right). \tag{22}$$

where $H(\cdot)$ denotes the Heaviside step function. Substituting (20)–(22) into (19), $\mathscr{Y}_i$ is obtained as

$$\mathscr{Y}_i = \frac{\tau}{2} e^{\omega_i} \int_0^{\infty} G_{0,1}^{1,0}\left(c_i \tau t \,\middle|\, \begin{matrix} - \\ 0 \end{matrix}\right) \times G_{0,2}^{2,0}\left(\lambda_{\mathrm{SBD}} \tau t \,\middle|\, \begin{matrix} - \\ 0,0 \end{matrix}\right) \times G_{1,1}^{0,1}\left(t \,\middle|\, \begin{matrix} 1 \\ 0 \end{matrix}\right) dt. \tag{23}$$

The integral in (23) is a Mellin-type integral[42] involving the product of three Meijer-G functions. By applying the generalized Fox-H function[43] identity for the product of Meijer-G functions, $\mathscr{Y}_i$ can be expressed as

$$\mathscr{Y}_i = \frac{1}{2c_i} e^{\omega_i} \mathscr{H}\left(\frac{\lambda_{\mathrm{SBD}}}{c_i}, \frac{1}{\tau c_i}\right), \qquad i \in \{1,2\}. \tag{24}$$

For completeness, the generalized H-function framework underlying the representation in (24) is discussed in[44]. The bivariate generalized Fox-H function used herein is defined as

$$\mathscr{H}(u,v) \triangleq H_{1,0:0,2:1,1}^{0,1:2,0:0,1}\left[\begin{matrix} (0;1,1) \\ - \end{matrix} \,\middle|\, \begin{matrix} - \\ (0,1),(0,1) \end{matrix} \,\middle|\, \begin{matrix} (1,1) \\ (0,1) \end{matrix} \,\middle|\, u,v\right]. \tag{25}$$

For the parameter set specified in (25), the generalized Fox–H function admits the equivalent real-integral representation $\mathscr{H}(u,v) = 2\int_{1/v}^{\infty} e^{-t} K_0\left(2\sqrt{ut}\right) \mathrm{d}t$, $u, v > 0$. This identity is obtained by applying the substitution $t = c_i x$ to the integral defining Y_i in (17) and comparing the result with its generalized Fox–H representation in (24). The same representation is also used to evaluate the IP expression in (37).

Finally, substituting (24) into (17), the closed-form OP is given by

$$\mathrm{OP} = 1 - \frac{\lambda_{\mathrm{SBD}} \lambda_{\mathrm{ED}}}{\mathscr{A}} \left[\frac{1}{c_1} \exp\left(-\frac{\lambda_{\mathrm{SD}} \gamma_{\mathrm{th}}}{\rho_{\mathrm{S}}}\right) \mathscr{H}\left(\frac{\lambda_{\mathrm{SBD}}}{c_1}, \frac{1}{\tau c_1}\right) - \frac{1}{c_2} \exp\left(\frac{\lambda_{\mathrm{ED}}}{\rho_{\mathrm{E}}}\right) \mathscr{H}\left(\frac{\lambda_{\mathrm{SBD}}}{c_2}, \frac{1}{\tau c_2}\right)\right]. \tag{26}$$

Equation (26) provides an explicit closed-form expression for the OP in terms of the bivariate generalized Fox-H function.

3.2 High-SNR Approximation of the Outage Probability

To gain further insight into the limiting reliability performance, we consider the high-SNR regime in which $\rho_{\mathrm{S}} \to \infty$ while ρ_{E} remains fixed. In this regime, $\tau = \frac{\widetilde{\gamma}_{\mathrm{th}}}{\beta\rho_{\mathrm{S}}} \to 0$, $\frac{\lambda_{\mathrm{ED}}}{\mathscr{A}} \to 1$, $\exp\left(-\frac{\lambda_{\mathrm{SD}}\gamma_{\mathrm{th}}}{\rho_{\mathrm{S}}}\right) \to 1$. Furthermore, $c_2 = \lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}\left(1+\frac{1}{\widetilde{\gamma}_{\mathrm{th}}}\right) + \frac{\lambda_{\mathrm{ED}}\beta\rho_{\mathrm{S}}}{\widetilde{\gamma}_{\mathrm{th}}\rho_{\mathrm{E}}} \to \infty$, such that the second integral term in (16) vanishes asymptotically. Therefore, the asymptotic OP is obtained as

$$\mathrm{OP}_{\infty} = 1 - \int_0^{\infty} e^{-\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}x} 2\lambda_{\mathrm{SBD}} K_0\left(2\sqrt{\lambda_{\mathrm{SBD}}x}\right) dx. \tag{27}$$

The integral in (27) can be evaluated by invoking the following standard Laplace-transform identity for the modified Bessel function of the second kind[45]:

$$\int_0^{\infty} e^{-\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}x} 2\lambda_{\mathrm{SBD}} K_0\left(2\sqrt{\lambda_{\mathrm{SBD}}x}\right) dx = \frac{\lambda_{\mathrm{SBD}}}{\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}} \exp\left(\frac{\lambda_{\mathrm{SBD}}}{\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}}\right) E_1\left(\frac{\lambda_{\mathrm{SBD}}}{\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}}\right). \tag{28}$$

where $E_1(\cdot)$ denotes the exponential integral function. Hence,

$$\mathrm{OP}_{\infty} = 1 - \frac{\lambda_{\mathrm{SBD}}}{\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}} \exp\left(\frac{\lambda_{\mathrm{SBD}}}{\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}}\right) E_1\left(\frac{\lambda_{\mathrm{SBD}}}{\lambda_{\mathrm{SD}}\beta\gamma_{\mathrm{th}}}\right). \tag{29}$$

Under the considered high-SNR regime with fixed ρ_E, (29) shows that the asymptotic OP does not explicitly depend on either ρ_S or ρ_E. As $\rho_S \to \infty$ with fixed ρ_E, the second-stage SINR in (5) grows without bound almost surely, whereas the first-stage SINR in (4) converges to an interference-limited quantity determined by the direct and backscatter channel gains. Consequently, the outage floor is governed by the source-decoding threshold γ_{th}, while the backscatter-decoding threshold $\widetilde{\gamma}_{\mathrm{th}}$ affects only the finite-SNR performance. Therefore, increasing the source transmit power alone cannot eliminate the outage floor.

It should be emphasised that this conclusion applies to a fixed and finite ρ_{E}. If the eavesdropper power scales with ρ_{S}, the relative scaling rate determines whether the same outage floor remains valid. A joint discussion of the corresponding source–eavesdropper power-scaling regimes is provided after (40).

4 Security Analysis

4.1 Intercept Probability Analysis

In this section, we characterize the security performance of the considered system from the eavesdropper's perspective. Because the source and backscatter transmissions convey complementary components of the same protected content unit, complete interception occurs only when E successfully recovers both components. The IP is therefore defined as the joint probability that the source- and backscatter-decoding requirements are simultaneously satisfied at E. In the considered SIC-based BC system, successful interception requires that the AE E successfully decodes both the source signal and the backscatter signal. Specifically, the source signal is first decoded while treating the backscatter signal and the residual self-interference as interference. After successful decoding, perfect SIC is assumed, allowing the source component to be completely removed before decoding the backscatter signal. Accordingly, the IP is expressed as

$$\begin{aligned} \mathrm{IP} &= \Pr\left(\mathrm{SINR}_{\mathrm{E},1} \geq \gamma_{\mathrm{th}}, \mathrm{SINR}_{\mathrm{E},2} \geq \widetilde{\gamma}_{\mathrm{th}}\right) \\ &= \Pr\left(\frac{\rho_{\mathrm{S}}\gamma_{\mathrm{SE}}}{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBE}} + \rho_{\mathrm{E}}\gamma_{\mathrm{EE}} + 1} \geq \gamma_{\mathrm{th}}, \frac{\beta\rho_{\mathrm{S}}\gamma_{\mathrm{SBE}}}{\rho_{\mathrm{E}}\gamma_{\mathrm{EE}} + 1} \geq \widetilde{\gamma}_{\mathrm{th}}\right). \end{aligned} \tag{30}$$

Accordingly, the IP considered in this work quantifies complete-content interception rather than successful recovery of either individual information component. It is worth noting that the IP characterizes successful complete-content recovery at the eavesdropper, whereas the OP characterizes the failure of complete recovery at the legitimate destination. Owing to the analogous SIC structures at D and E, the IP can be derived following the same conditioning procedure as that used for the OP, with the parameter mapping $\{\lambda_{\mathrm{SD}}, \lambda_{\mathrm{ED}}, \lambda_{\mathrm{SBD}}\} \to \{\lambda_{\mathrm{SE}}, \lambda_{\mathrm{EE}}, \lambda_{\mathrm{SBE}}\}$. Let $x = \gamma_{\mathrm{SBE}}$. The second-stage decoding condition in (30) yields the upper bound for γ_{EE}:

$$\gamma_{\mathrm{EE}} \leq \xi_{\mathrm{E}} = \frac{\beta\rho_{\mathrm{S}}x - \widetilde{\gamma}_{\mathrm{th}}}{\widetilde{\gamma}_{\mathrm{th}}\rho_{\mathrm{E}}}. \tag{31}$$

Since γ_{EE} is a non-negative random variable, the feasible decoding region exists only when $\xi_{\mathrm{E}} \geq 0$, which is equivalent to $x \geq \tau_{\mathrm{E}}$, where $\tau_{\mathrm{E}} = \frac{\widetilde{\gamma}_{\mathrm{th}}}{\beta \rho_{\mathrm{S}}}$. Conditioned on the cascaded channel gain $\gamma_{\mathrm{SBE}} = x$, the IP can be obtained by averaging over the distribution of the residual self-interference channel gain γ_{EE}. Following the same conditioning procedure as in the OP analysis, the exact IP is expressed as

$$\mathrm{IP} = \frac{\lambda_{\mathrm{EE}}}{\lambda_{\mathrm{EE}} + \lambda_{\mathrm{SE}} \gamma_{\mathrm{th}} \frac{\rho_{\mathrm{E}}}{\rho_{\mathrm{S}}}} \left[e^{-\frac{\lambda_{\mathrm{SE}} \gamma_{\mathrm{th}}}{\rho_{\mathrm{S}}}} \int_{\tau_{\mathrm{E}}}^{\infty} e^{-\lambda_{\mathrm{SE}} \beta \gamma_{\mathrm{th}} x} f_{\gamma_{\mathrm{SBE}}}(x)\, dx - e^{\frac{\lambda_{\mathrm{EE}}}{\rho_{\mathrm{E}}}} \int_{\tau_{\mathrm{E}}}^{\infty} e^{-\left[\lambda_{\mathrm{SE}} \beta \gamma_{\mathrm{th}} \left(1 + \frac{1}{\widetilde{\gamma}_{\mathrm{th}}}\right) + \frac{\lambda_{\mathrm{EE}} \beta \rho_{\mathrm{S}}}{\widetilde{\gamma}_{\mathrm{th}} \rho_{\mathrm{E}}}\right] x} f_{\gamma_{\mathrm{SBE}}}(x)\, dx \right]. \tag{32}$$

Since γ_{SB} and γ_{BE} are independent exponentially distributed random variables, the PDF of the cascaded channel gain γ_{SBE} is given by

$$f_{\gamma_{\mathrm{SBE}}}(x) = 2\lambda_{\mathrm{SBE}} K_0 \left(2\sqrt{\lambda_{\mathrm{SBE}} x}\right). \tag{33}$$

For notational compactness, define $\mathscr{A}_{\mathrm{E}} \triangleq \lambda_{\mathrm{EE}} + \lambda_{\mathrm{SE}} \gamma_{\mathrm{th}} \frac{\rho_{\mathrm{E}}}{\rho_{\mathrm{S}}}$, $c_{\mathrm{E},1} \triangleq \lambda_{\mathrm{SE}} \beta \gamma_{\mathrm{th}}$, $c_{\mathrm{E},2} \triangleq \lambda_{\mathrm{SE}} \beta \gamma_{\mathrm{th}} \left(1 + \frac{1}{\widetilde{\gamma}_{\mathrm{th}}}\right) + \frac{\lambda_{\mathrm{EE}} \beta \rho_{\mathrm{S}}}{\widetilde{\gamma}_{\mathrm{th}} \rho_{\mathrm{E}}}$. Then, (32) becomes

$$\mathrm{IP} = \frac{2\lambda_{\mathrm{SBE}} \lambda_{\mathrm{EE}}}{\mathscr{A}_{\mathrm{E}}} (z_1 - z_2), \tag{34}$$

where

$$z_i = e^{\omega_{\mathrm{E},i}} \int_{\tau_{\mathrm{E}}}^{\infty} e^{-c_{\mathrm{E},i} x} K_0 \left(2\sqrt{\lambda_{\mathrm{SBE}} x}\right) dx, \qquad i \in \{1,2\}, \tag{35}$$

with $\omega_{\mathrm{E},1} = -\frac{\lambda_{\mathrm{SE}} \gamma_{\mathrm{th}}}{\rho_{\mathrm{S}}}, \omega_{\mathrm{E},2} = \frac{\lambda_{\mathrm{EE}}}{\rho_{\mathrm{E}}}$. Following exactly the same derivation as in the OP analysis, the integral in (35) admits a generalized Fox-H[43] representation. Consequently, z_i is obtained as

$$z_i = \frac{1}{2c_{\mathrm{E},i}} e^{\omega_{\mathrm{E},i}} \mathscr{H}\left(\frac{\lambda_{\mathrm{SBE}}}{c_{\mathrm{E},i}}, \frac{1}{\tau_{\mathrm{E}} c_{\mathrm{E},i}}\right), \qquad i \in \{1,2\}, \tag{36}$$

where $\mathscr{H}(\cdot,\cdot)$ denotes the bivariate generalized Fox-H function defined in (25). Substituting (36) into (34), the closed-form IP is given by

$$\mathrm{IP} = \frac{\lambda_{\mathrm{SBE}} \lambda_{\mathrm{EE}}}{A_{\mathrm{E}}} \left[\frac{1}{c_{\mathrm{E},1}} \exp\left(-\frac{\lambda_{\mathrm{SE}} \gamma_{\mathrm{th}}}{\rho_{\mathrm{S}}}\right) \mathscr{H}\left(\frac{\lambda_{\mathrm{SBE}}}{c_{\mathrm{E},1}}, \frac{1}{\tau_{\mathrm{E}} c_{\mathrm{E},1}}\right) - \frac{1}{c_{\mathrm{E},2}} \exp\left(\frac{\lambda_{\mathrm{EE}}}{\rho_{\mathrm{E}}}\right) \mathscr{H}\left(\frac{\lambda_{\mathrm{SBE}}}{c_{\mathrm{E},2}}, \frac{1}{\tau_{\mathrm{E}} c_{\mathrm{E},2}}\right) \right]. \tag{37}$$

Equation (37) reveals that the interception capability depends jointly on the quality of the source–eavesdropper link, the cascaded backscatter channel, the residual self-interference, and the artificial jamming power. In particular, stronger residual self-interference or a larger normalized eavesdropper jamming power ρ_{E} degrades the decoding capability of the AE, thereby reducing the IP and improving the PLS performance. Since the active FD eavesdropper controls its own jamming power, the preferred value of ρ_{E} depends on the attack objective. For a given channel realization, increasing ρ_{E} reduces both $\mathrm{SINR}_{\mathrm{D},1}$ and $\mathrm{SINR}_{\mathrm{D},2}$ through the additional interference term $\rho_{\mathrm{E}} \gamma_{\mathrm{ED}}$, whereas it also reduces $\mathrm{SINR}_{\mathrm{E},1}$ and $\mathrm{SINR}_{\mathrm{E},2}$ by increasing the residual self-interference term $\rho_{\mathrm{E}} \gamma_{\mathrm{EE}}$. Consequently, there is no single jamming-power setting that is simultaneously most favorable for both interception and disruption. Within a prescribed admissible jamming-power range, an interception-oriented adversary favors the lower boundary because reducing ρ_{E} alleviates its residual self-interference. If zero jamming power is admissible, $\rho_{\mathrm{E}} = 0$ corresponds to the passive-eavesdropping boundary. In contrast, a disruption-oriented adversary favors the upper boundary because increasing ρ_{E} degrades both decoding stages at the legitimate destination. For a dual-purpose adversary, these two objectives are conflicting, and intermediate jamming levels therefore represent different operating points between interception capability and disruption effectiveness. This objective-dependent behavior is consistent with the numerical results in Fig. 4 and Fig. 5, where the OP increases whereas the IP decreases monotonically with ρ_{E}.

4.2 High-SNR Approximation of the Intercept Probability

To characterize the asymptotic interception behavior, we consider $\rho_{\mathrm{S}} \to \infty$ while keeping ρ_{E} fixed. In this regime, $\tau_{\mathrm{E}} \to 0, \frac{\lambda_{\mathrm{EE}}}{A_{\mathrm{E}}} \to 1, \exp\left(-\frac{\lambda_{\mathrm{SE}} \gamma_{\mathrm{th}}}{\rho_{\mathrm{S}}}\right) \to 1$.

Moreover,

$$c_{\mathrm{E},2} = \lambda_{\mathrm{SE}}\beta\gamma_{\mathrm{th}}\left(1+\frac{1}{\widetilde{\gamma}_{\mathrm{th}}}\right)+\frac{\lambda_{\mathrm{EE}}\beta\rho_{\mathrm{S}}}{\widetilde{\gamma}_{\mathrm{th}}\rho_{\mathrm{E}}} \to \infty, \tag{38}$$

so that the second integral term in (32) vanishes asymptotically.

Therefore, the asymptotic IP is obtained as

$$\mathrm{IP}_{\infty} = \int_0^{\infty} e^{-\lambda_{\mathrm{SE}}\beta\gamma_{\mathrm{th}}x} 2\lambda_{\mathrm{SBE}} K_0\left(2\sqrt{\lambda_{\mathrm{SBE}}x}\right) dx. \tag{39}$$

Using the standard Laplace-transform[45] identity of the modified Bessel function, (39) yields

$$\mathrm{IP}_{\infty} = \frac{\lambda_{\mathrm{SBE}}}{\lambda_{\mathrm{SE}}\beta\gamma_{\mathrm{th}}} \exp\left(\frac{\lambda_{\mathrm{SBE}}}{\lambda_{\mathrm{SE}}\beta\gamma_{\mathrm{th}}}\right) E_1\left(\frac{\lambda_{\mathrm{SBE}}}{\lambda_{\mathrm{SE}}\beta\gamma_{\mathrm{th}}}\right). \tag{40}$$

Remark 1: Limitation of the fixed-jamming-power asymptotic regime.

The asymptotic expressions in (29) and (40) are conditional on $\rho_{\mathrm{S}} \to \infty$ while ρ_{E} remains fixed and finite. This assumption represents a power-limited active eavesdropper whose jamming power is determined independently of the source transmit power. It is relevant when the eavesdropper is subject to a fixed hardware, energy, or regulatory power constraint. However, it does not describe an adaptive adversary whose jamming power increases with ρ_{S}. Consequently, the independence of the asymptotic OP and IP from ρ_{E} should not be interpreted as a general result for arbitrary power-scaling policies. To illustrate this limitation, consider the more general scaling model

$$\rho_{\mathrm{E}} = \kappa\rho_{\mathrm{S}}^{\alpha}, \qquad \kappa > 0, \quad \alpha \geq 0. \tag{41}$$

For $0 \leq \alpha < 1$, the eavesdropper power grows more slowly than the source power. The jamming and residual self-interference terms are then asymptotically dominated by the source-powered terms, and the limits in (29) and (40) remain valid. For $\alpha = 1$, the source and eavesdropper powers scale proportionally. The limiting SINRs become

$$\begin{aligned} \mathrm{SINR}_{\mathrm{D},1}^{(\infty)} &= \frac{\gamma_{\mathrm{SD}}}{\beta\gamma_{\mathrm{SBD}}+\kappa\gamma_{\mathrm{ED}}}, \quad \mathrm{SINR}_{\mathrm{D},2}^{(\infty)} = \frac{\beta\gamma_{\mathrm{SBD}}}{\kappa\gamma_{\mathrm{ED}}}, \\ \mathrm{SINR}_{\mathrm{E},1}^{(\infty)} &= \frac{\gamma_{\mathrm{SE}}}{\beta\gamma_{\mathrm{SBE}}+\kappa\gamma_{\mathrm{EE}}}, \quad \mathrm{SINR}_{\mathrm{E},2}^{(\infty)} = \frac{\beta\gamma_{\mathrm{SBE}}}{\kappa\gamma_{\mathrm{EE}}}. \end{aligned} \tag{42}$$

Therefore, both SIC stages remain limiting, and the asymptotic OP and IP depend on κ, the jamming and residual self-interference channels, and both decoding thresholds γ_{th} and $\widetilde{\gamma}_{\mathrm{th}}$. If $\alpha > 1$, the eavesdropper power grows faster than the source power. Under the adopted residual self-interference model, all four decoding SINRs tend to zero almost surely, giving

$$\mathrm{OP} \to 1, \qquad \mathrm{IP} \to 0. \tag{43}$$

For fixed, nonzero jammer-to-BD coupling, the $\alpha > 1$ case in (43) is a formal limit of the reduced signal model, since the source-dominated illumination condition in (3) cannot be maintained as $\rho_{\mathrm{S}} \to \infty$. Hence, (29) and (40) should be viewed as source-power-dominant asymptotic benchmarks rather than universal high-SNR limits. A complete analysis under jointly scaling or adaptively controlled source and eavesdropper powers is beyond the scope of this work and constitutes an important direction for future study.

4.3 Optimal Backscatter Coefficient

As shown in the preceding analysis, the backscatter coefficient β has a non-monotonic impact on the reliability and interception performance. Therefore, the reliability-oriented optimal backscatter coefficient is defined as

$$\beta_{\mathrm{OP}}^{\star} = \arg\min_{0<\beta<1} \mathrm{OP}(\beta), \tag{44}$$

whereas the interception-critical operating point is defined as

$$\beta_{\mathrm{IP}}^{\star} = \arg\max_{0<\beta<1} \mathrm{IP}(\beta). \tag{45}$$

These two operating points characterize the operating conditions associated with favorable reliability and critical interception within the considered security and reliability trade-off. Owing to the mathematical structure of the derived OP and IP expressions, obtaining $\beta^{\star}_{\mathrm{OP}}$ and $\beta^{\star}_{\mathrm{IP}}$ in closed form is analytically intractable. Hence, following the scalar optimization approach in[46], the Golden-section search (GSS) method is employed to numerically determine the corresponding optimal backscatter coefficient over the feasible interval of β. A detailed description of the algorithm and its theoretical foundation is provided in[47]. The same GSS-based optimization is applied under both perfect and imperfect CSI conditions, and the resulting optimal values are investigated in Section 5.

5 Numerical Results and Discussion

This section examines the effects of the normalized source transmit power ρ_{S}, the normalized eavesdropper jamming power ρ_{E}, the backscatter coefficient β, and the target rates R_{S} and R_{B} on the OP and IP. Unless otherwise stated, the parameters are listed in Table 2. The analytical results, labeled "Ana", are obtained from (26) and (37). The Fox–H function in (25) is evaluated through its equivalent real-integral representation using MATLAB's integral routine for adaptive quadrature[48], with requested relative and absolute tolerances of 10^{-9} and 10^{-12}, respectively. Numerical consistency of the closed-form OP and IP expressions is further checked against direct quadrature of (15) and (32), respectively, using requested relative and absolute tolerances of 10^{-8} and 10^{-12}. The Monte Carlo results, labeled "Sim", use 10^6 independent channel realizations for each probability estimate. The elementary channel power gains are generated according to Section 2 as exponential random variables with means $1/\lambda_i$, and the cascaded gains are formed as products of the corresponding hop gains. Both SIC-stage conditions at each receiver are evaluated jointly within the same realization, preserving the dependence specified by the decoding events in (8) and (30). The OP is estimated as the fraction of realizations with at least one decoding failure at D, whereas the IP is estimated as the fraction with both decoding conditions satisfied at E. The finite power sweeps are interpreted under the illumination and reflected-jamming conditions stated with (3), throughout the considered parameter range. In particular, source-dominated illumination requires sufficiently weak jammer-to-BD coupling at the largest considered ratio $\rho_{\mathrm{E}}/\rho_{\mathrm{S}}$. These conditions concern received powers and do not require $P_{\mathrm{E}} < P_{\mathrm{S}}$. Both the analytical and Monte Carlo evaluations omit the same jammer-originated reflected components while retaining the source-excited backscatter signals.

Table 2. Simulation Parameters.

Parameter	Symbol	Value
S→BD channel parameter	λ_{SB}	0.8
S→D channel parameter	λ_{SD}	0.4
S→E channel parameter	λ_{SE}	0.4
BD→D channel parameter	λ_{BD}	0.6
BD→E channel parameter	λ_{BE}	0.6
E→D channel parameter	λ_{ED}	1.5
Self-interference channel parameter	λ_{EE}	1.5
Backscatter coefficient	β	0.6
Source target rate	R_{S}	0.5 bit/s/Hz
Backscatter target rate	R_{B}	0.5 bit/s/Hz
Normalized eavesdropper jamming power	ρ_{E}	-10 to 30 dB
Normalized source transmit power	ρ_{S}	-10 to 30 dB
Monte Carlo iterations	N	10^6

Fig. 2 depicts the OP versus the backscatter coefficient β for different normalized source transmit powers, i.e., $\rho_{\mathrm{S}} \in \{5, 10, 20\}$ dB. The close match between the analytical and Monte Carlo simulation results confirms the correctness of the derived OP expression. It can be observed that increasing ρ_{S} generally reduces the OP, since a larger source power improves both the direct and cascaded backscatter links. For $\rho_{\mathrm{S}} = 10$ and 20 dB, the OP exhibits a clear non-monotonic behavior with respect to β. When β is small, only a limited portion of the incident signal is reflected by the BD, resulting in a weak cascaded backscatter signal and hence a high OP. As β increases, the useful backscattered signal becomes stronger, which improves the second-stage SIC decoding and significantly decreases the OP. However, when β becomes excessively large, the reflected signal also acts as co-channel interference during the first SIC stage, as shown in the denominator of $\mathrm{SINR}_{\mathrm{D},1}$. Consequently, the OP starts to increase again, especially in the high-ρ_{S} regime. In contrast, for the low-power case $\rho_{\mathrm{S}} = 5$ dB, the OP decreases with β and then approaches a nearly flat floor, indicating that the system performance is mainly limited by the insufficient source

power rather than by the backscatter-induced interference. Moreover, the OP-minimizing value of β shifts to a smaller region as ρ_S increases, suggesting that a high source power requires a more conservative reflection coefficient to avoid excessive interference in the first SIC stage. These observations highlight the importance of properly tuning β to balance backscatter signal enhancement and SIC-stage interference in secure source-assisted BC systems.

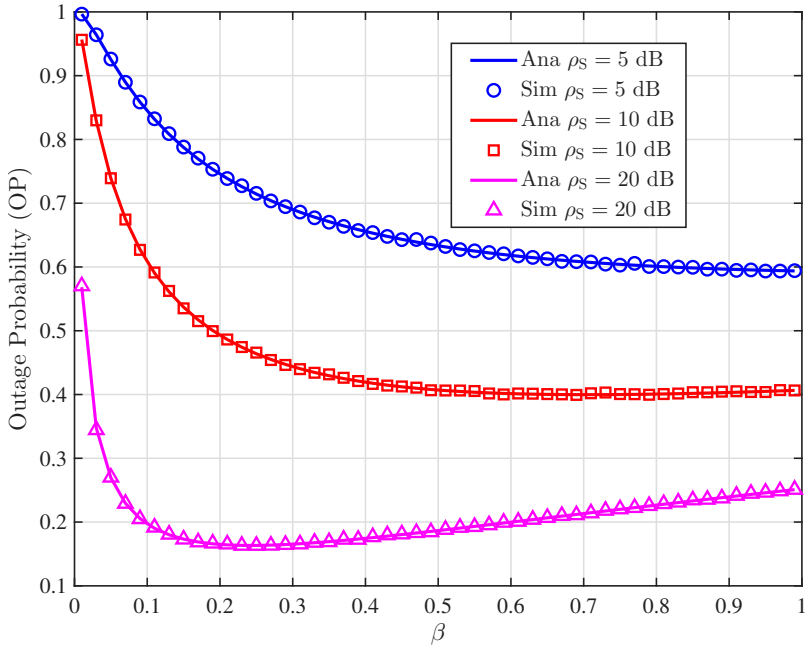


Figure 2. OP versus the backscatter coefficient β under different normalized source transmit powers ρ_S.

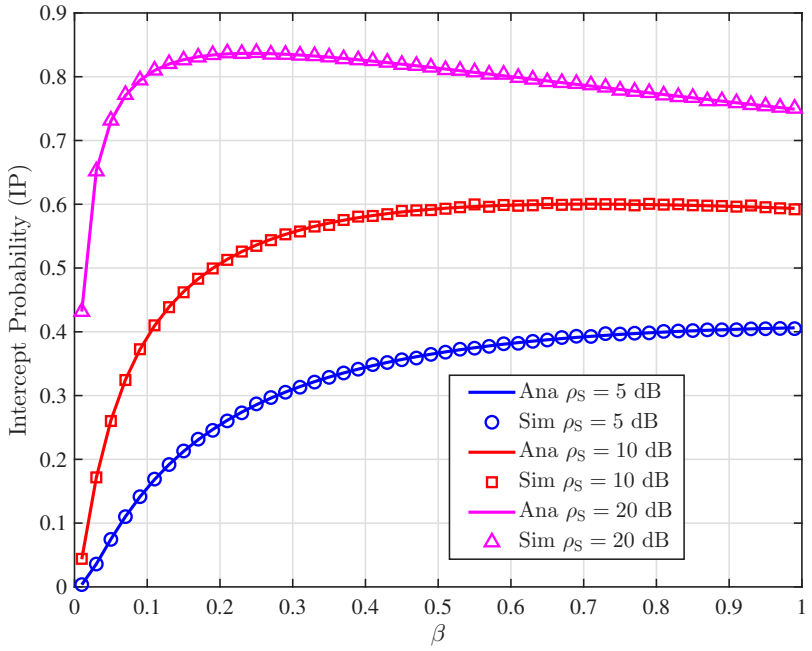


Figure 3. IP versus the backscatter coefficient β under different normalized source transmit powers ρ_S.

Fig. 3 shows the IP versus the backscatter coefficient β for different normalized source transmit powers, i.e., $\rho_S \in \{5, 10, 20\}$ dB. The analytical curves closely follow the Monte Carlo simulation results over the entire range of β, which further verifies the accuracy of the derived IP expression. It is observed that increasing ρ_S significantly increases the IP. This is because a larger source transmit power strengthens the signals received at the eavesdropper, thereby improving its probability of successfully

decoding the confidential information. For a given ρ_S, the IP exhibits a non-monotonic behavior with respect to β. In the low-β region, increasing β enhances the backscattered signal received at E, leading to a rapid increase in the IP. However, when β becomes large, the reflected signal also contributes stronger co-channel interference during the first SIC stage at the eavesdropper, as indicated by the denominator of $\mathrm{SINR}_{E,1}$. As a result, the IP tends to saturate or slightly decrease in the high-β region. This effect is particularly evident for $\rho_S = 20$ dB, where the IP reaches its maximum at a relatively small value of β and then gradually decreases as β further increases. Comparing Fig. 2 and Fig. 3, one can observe a clear reliability-security trade-off. A higher source transmit power reduces the OP at the legitimate destination, but it also increases the IP at the eavesdropper. Therefore, simply increasing ρ_S improves reliability at the cost of weakened PLS. These results indicate that the backscatter coefficient should be properly selected according to both reliability and security requirements, rather than being maximized indiscriminately.

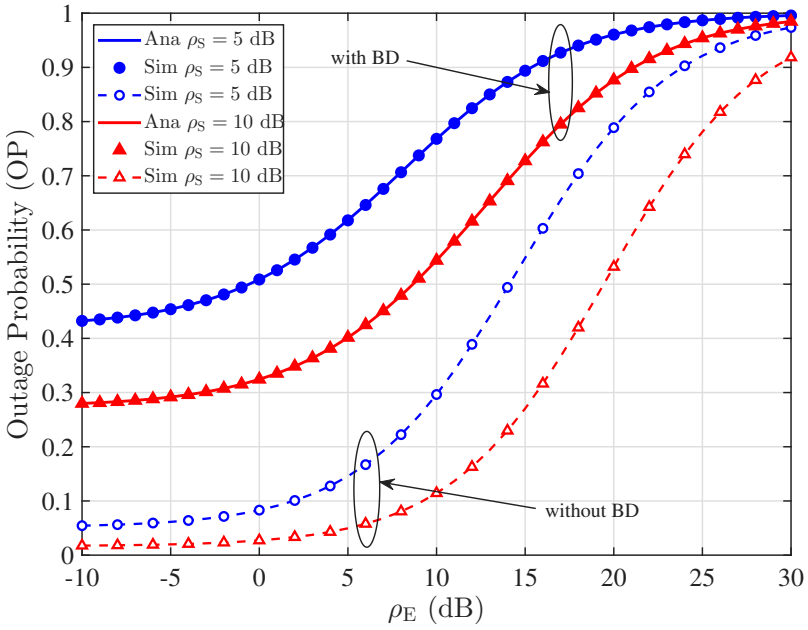


Figure 4. OP versus ρ_E for the proposed scheme and the direct-transmission benchmark without the BD, with $\rho_S \in \{5, 10\}$ dB, and $R_S = R_B = 0.5$ bit/s/Hz for the proposed scheme.

Fig. 4 shows the OP versus the normalized eavesdropper jamming power ρ_E for $\rho_S \in \{5, 10\}$ dB. For the proposed backscatter-assisted scheme, the analytical curves closely match the Monte Carlo results, further confirming the accuracy of the derived OP expression. As ρ_E increases, the OP rises monotonically because the jamming term $\rho_E \gamma_{ED}$ degrades the decoding conditions at D in both SIC stages, as shown in (4) and (5). Increasing ρ_S from 5 to 10 dB improves reliability by strengthening the desired source and backscatter components relative to the jamming and noise terms. The direct-transmission benchmark exhibits a markedly lower OP than the proposed scheme at the same ρ_S and ρ_E. When the BD is removed, only the source signal remains to be decoded and the received SINR at D reduces to $\mathrm{SINR}_D^{\mathrm{dir}} = \frac{\rho_S \gamma_{SD}}{\rho_E \gamma_{ED}+1}$. Compared with $\mathrm{SINR}_{D,1}$ in (4), this expression no longer contains the backscatter-induced interference term $\beta\rho_S\gamma_{SBD}$, and no second-stage backscatter decoding condition is required. Hence, successful reception in the direct-transmission benchmark depends only on whether $\mathrm{SINR}_D^{\mathrm{dir}}$ satisfies the source-decoding threshold γ_{th}. This explains the lower OP of the without-BD curves and shows that the proposed architecture incurs an additional reliability cost because both information components must be successfully recovered. The benefit of removing the BD from the reliability perspective is therefore accompanied by a fundamentally different interception condition at the eavesdropper.

The corresponding IP results are presented in Fig. 5. The analytical curves for the proposed scheme remain in close agreement with the Monte Carlo simulations over the considered range of ρ_E. In contrast to the OP behavior in Fig. 4, the IP decreases monotonically as ρ_E increases. This behavior follows directly from the SINRs in (6) and (7), where a larger jamming level strengthens the residual self-interference term $\rho_E \gamma_{EE}$ and thereby degrades the decoding capability of the FD eavesdropper. Conversely, increasing ρ_S from 5 to 10 dB raises the IP because both information-bearing components are received more strongly at E. For the direct-transmission benchmark, removing the BD leaves only the source component to be recovered at the eavesdropper, yielding $\mathrm{SINR}_E^{\mathrm{dir}} = \frac{\rho_S \gamma_{SE}}{\rho_E \gamma_{EE}+1}$. This expression shows that the backscatter-induced interference term $\beta\rho_S\gamma_{SBE}$ is absent and, more importantly, that interception no longer depends on successful recovery of a second information component.

Hence, the direct-transmission benchmark requires only $\mathrm{SINR}_{\mathrm{E}}^{\mathrm{dir}} \geq \gamma_{\mathrm{th}}$, whereas complete interception in the proposed scheme requires both source and backscatter decoding conditions to be satisfied. This difference in the decoding requirements explains why the without-BD benchmark exhibits a substantially higher IP for the same ρ_{S} and ρ_{E}. The separation is particularly evident in the low and moderate-jamming regions, where the eavesdropper retains sufficient reception capability to benefit from the simpler single-component decoding requirement. As ρ_{E} increases further, residual self-interference progressively dominates the eavesdropper reception in both transmission schemes, driving their IPs toward low values and reducing the difference between them.

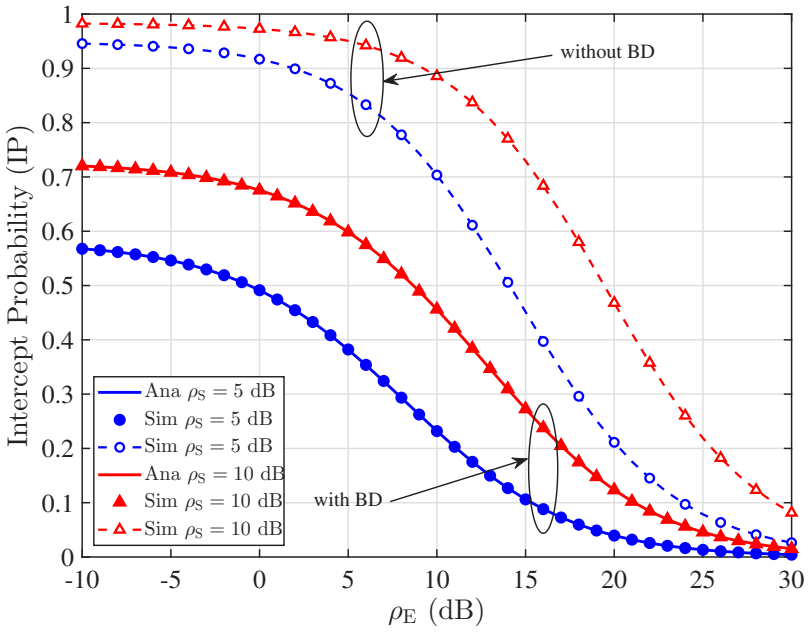


Figure 5. IP versus ρ_{E} for the proposed scheme and the direct-transmission benchmark without the BD, with $\rho_{\mathrm{S}} \in \{5, 10\}$ dB, and $R_{\mathrm{S}} = R_{\mathrm{B}} = 0.5$ bit/s/Hz for the proposed scheme.

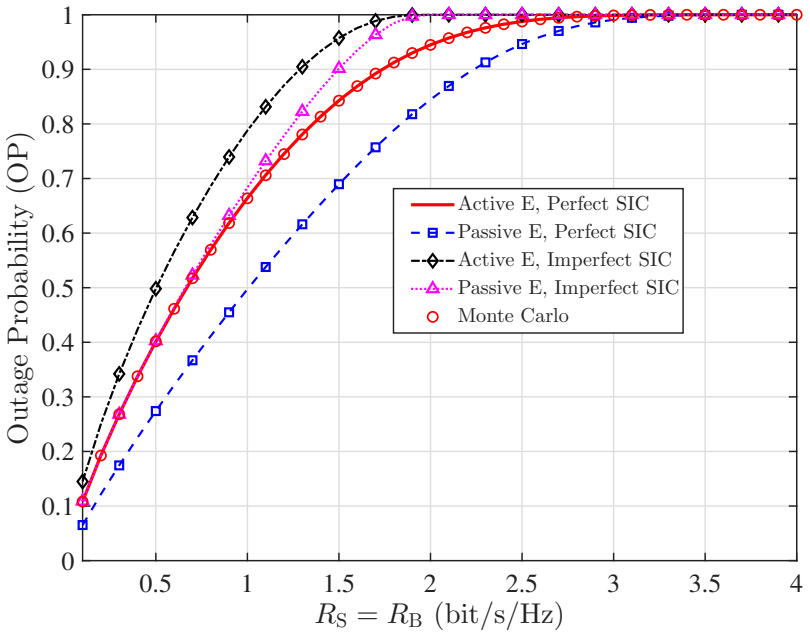


Figure 6. OP versus the target rate R under active and passive eavesdropping with perfect and imperfect SIC, with $\rho_{\mathrm{S}} = 10$ dB, $\rho_{\mathrm{E}} = 5$ dB for active E, $P_{\mathrm{E}} = 0$ for passive E, and $\omega = 0.1$ for imperfect SIC.

Fig. 6 compares the OP versus the common target rate R, with $R_S = R_B$, under active and passive eavesdropping with perfect and imperfect SIC. For the passive-eavesdropping benchmark, E operates only as an information receiver and does not transmit the jamming waveform. Accordingly, $P_E = 0$, or equivalently $\rho_E = 0$, so that the E$\to$D jamming component is absent. Under perfect SIC, the corresponding SINRs at D reduce to $\mathrm{SINR}_{D,1}^{pas} = \rho_S\gamma_{SD}/(\beta\rho_S\gamma_{SBD}+1)$ and $\mathrm{SINR}_{D,2}^{pas} = \beta\rho_S\gamma_{SBD}$. To examine the impact of imperfect SIC, we adopt the commonly used residual-interference model in[49], in which a fraction ω of the decoded source-signal power remains during the second decoding stage. A representative value of $\omega = 0.1$ is considered, consistent with the numerical imperfect-SIC setting in[49]. Since this residual component arises only after source decoding, the first-stage SINR remains unchanged, whereas the second-stage SINR for active E becomes $\mathrm{SINR}_{D,2}^{imSIC} = \beta\rho_S\gamma_{SBD}/(\rho_E\gamma_{ED}+\omega\rho_S\gamma_{SD}+1)$. For passive E, this expression further reduces to $\mathrm{SINR}_{D,2}^{pas,imSIC} = \beta\rho_S\gamma_{SBD}/(\omega\rho_S\gamma_{SD}+1)$. Based on these SINRs, the OP increases with R because the larger decoding thresholds make simultaneous recovery of the source and backscatter components more difficult. The passive-E case with perfect SIC yields the lowest OP because neither adversarial jamming nor residual post-cancellation source interference affects the second decoding stage. Activating the eavesdropper introduces the additional $\rho_E\gamma_{ED}$ interference term, which degrades the decoding conditions at D. Imperfect SIC causes a further reliability loss through $\omega\rho_S\gamma_{SD}$ in the second-stage SINR. Consequently, the active-E case with imperfect SIC exhibits the largest OP over the considered range. The passive-E case with imperfect SIC lies between the corresponding perfect-SIC benchmarks because residual SIC remains present while adversarial jamming is absent.

Fig. 7 presents the corresponding IP versus the common target rate R, with $R_S = R_B$, under the same active and passive eavesdropping conditions. For passive E, $P_E = 0$, or equivalently $\rho_E = 0$. Hence, the eavesdropper does not transmit a jamming waveform and does not experience the associated full-duplex residual self-interference. Under perfect SIC, its SINRs therefore reduce to $\mathrm{SINR}_{E,1}^{pas} = \rho_S\gamma_{SE}/(\beta\rho_S\gamma_{SBE}+1)$ and $\mathrm{SINR}_{E,2}^{pas} = \beta\rho_S\gamma_{SBE}$. Following the same imperfect-SIC model in[49] with $\omega = 0.1$, the first-stage SINR remains unchanged because residual cancellation arises only after successful source decoding. The second-stage SINR for active E becomes $\mathrm{SINR}_{E,2}^{imSIC} = \beta\rho_S\gamma_{SBE}/(\rho_E\gamma_{EE}+\omega\rho_S\gamma_{SE}+1)$, whereas for passive E it reduces to $\mathrm{SINR}_{E,2}^{pas,imSIC} = \beta\rho_S\gamma_{SBE}/(\omega\rho_S\gamma_{SE}+1)$. Based on these SINRs, the IP decreases with R because increasing the decoding thresholds makes simultaneous recovery of the source and backscatter components progressively more difficult at E. The passive-E case with perfect SIC yields the highest IP because neither the full-duplex residual self-interference term $\rho_E\gamma_{EE}$ nor the residual SIC term $\omega\rho_S\gamma_{SE}$ is present. Active eavesdropping lowers the IP through the residual self-interference generated by the eavesdropper's own transmission. Imperfect SIC further degrades second-stage backscatter decoding through $\omega\rho_S\gamma_{SE}$. Their combined effect explains why the active-E case with imperfect SIC yields the lowest IP among the considered configurations. Comparing Figs. 6 and 7 further shows that the adopted residual SIC model affects reliability and interception in opposite directions. It increases the OP at the legitimate destination while reducing the IP at the eavesdropper.

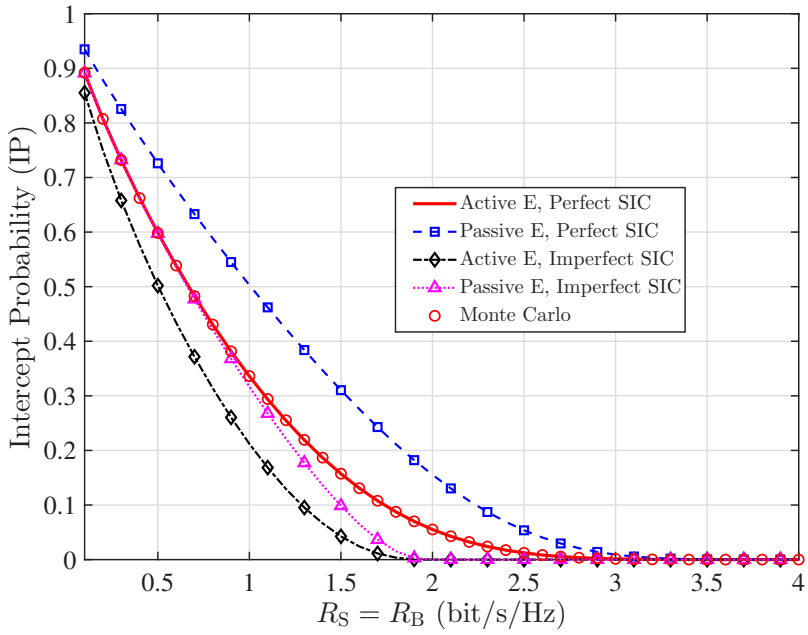


Figure 7. IP versus the target rate R under active and passive eavesdropping with perfect and imperfect SIC, with $\rho_S = 10$ dB, $\rho_E = 5$ dB for active E, $P_E = 0$ for passive E, and $\omega = 0.1$ for imperfect SIC.

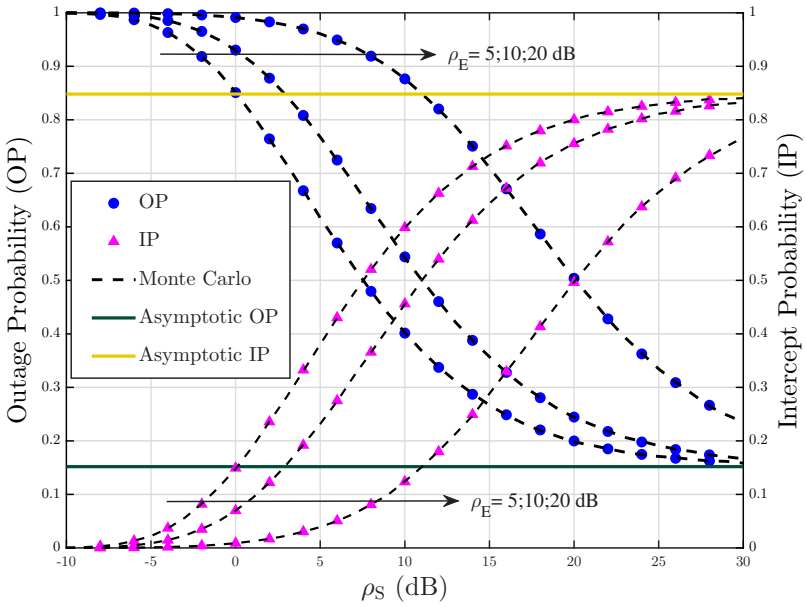


Figure 8. Reliability–security trade-off in terms of OP and IP versus ρ_S for $\rho_E \in \{5, 10, 20\}$ dB, with $\beta = 0.6$ and $R_S = R_B = 0.5$ bit/s/Hz.

Fig. 8 jointly illustrates the OP and IP versus the normalized source transmit power ρ_S for different normalized eavesdropper jamming powers, i.e., $\rho_E \in \{5, 10, 20\}$ dB. The OP and IP are shown with respect to the left and right vertical axes, respectively. The analytical results closely agree with the corresponding Monte Carlo simulations over the entire considered range of ρ_S, further validating the derived OP and IP expressions. The horizontal asymptotic lines also accurately capture the limiting values approached in the high-ρ_S regime. As ρ_S increases, the OP decreases whereas the IP increases, revealing a clear reliability-security trade-off. At sufficiently large ρ_S, both metrics approach non-zero asymptotic limits. As shown in (29) and (40), these limits are independent of the numerical value of ρ_E only under the adopted asymptotic regime in which $\rho_S \to \infty$ while ρ_E remains fixed and finite. In this source-power-dominant regime, the second SIC stage becomes asymptotically non-limiting, whereas the first stage remains limited by the backscatter component. The OP therefore approaches a non-zero floor, while the IP approaches a non-zero ceiling. This convergence should not be generalised to an adaptive eavesdropper whose jamming power scales proportionally to, or faster than, the source power. Under proportional scaling, the jamming and residual self-interference terms remain relevant in both SIC stages, and the limiting OP and IP depend on the source-to-jamming power ratio. The horizontal asymptotic lines in Fig. 8 should therefore be interpreted as benchmarks for a fixed-power active eavesdropper rather than as universal limits under arbitrary adversarial power control. Overall, Fig. 8 confirms that increasing the source transmit power alone cannot simultaneously improve reliability and security. Although a larger ρ_S reduces the OP, it also increases the IP until both metrics reach their respective interference-limited asymptotic values. Therefore, the source power should be selected jointly with the expected jamming level and the required outage and interception constraints when designing secure source-assisted BackCom systems.

Fig. 9 further examines the reliability and security tradeoff when distinct target rates are assigned to the source and backscatter signals. The analytical OP and IP results remain in close agreement with the corresponding Monte Carlo simulations for all considered (R_S, R_B) pairs, confirming the validity of the derived expressions beyond the equal rate setting. At finite ρ_S, increasing R_B from 0.5 to 1 bit/s/Hz while keeping $R_S = 0.5$ bit/s/Hz increases the OP and reduces the IP because the resulting increase in the backscatter decoding threshold $\widetilde{\gamma}_{th}$ makes the second SIC stage more difficult at both the legitimate destination and the eavesdropper. Consequently, a higher backscatter rate requirement imposes a reliability penalty at the legitimate destination while simultaneously making complete information recovery more difficult for the eavesdropper. In contrast, increasing R_S from 0.5 to 1 bit/s/Hz with $R_B = 0.5$ bit/s/Hz affects both the finite SNR behavior and the limiting performance. For the parameter setting of Fig. 9, the cases $(R_S, R_B) = (0.5, 0.5)$ and $(0.5, 1.0)$ converge to the same asymptotic values, namely $\text{OP}_\infty \approx 0.1519$ and $\text{IP}_\infty \approx 0.8481$. When R_S is increased to 1 bit/s/Hz, the corresponding limits become $\text{OP}_\infty \approx 0.2773$ and $\text{IP}_\infty \approx 0.7227$. These numerical values are fully consistent with (29) and (40), in which the asymptotic OP and IP depend on the source decoding threshold $\gamma_{th} = 2^{R_S} - 1$, whereas the backscatter decoding threshold $\widetilde{\gamma}_{th} = 2^{R_B} - 1$ does not appear. This occurs because, with fixed ρ_E, the second SIC stage becomes nonlimiting as ρ_S increases, while the first decoding stage remains interference limited by the backscatter component. Hence, R_B mainly controls the finite SNR decoding burden of the

second SIC stage, whereas R_S determines the high SNR outage floor and interception ceiling. The results therefore show that the equal rate configuration $R_S = R_B$ represents only a special operating case and that independently selecting the source and backscatter target rates can shift the system to substantially different reliability and security operating points.

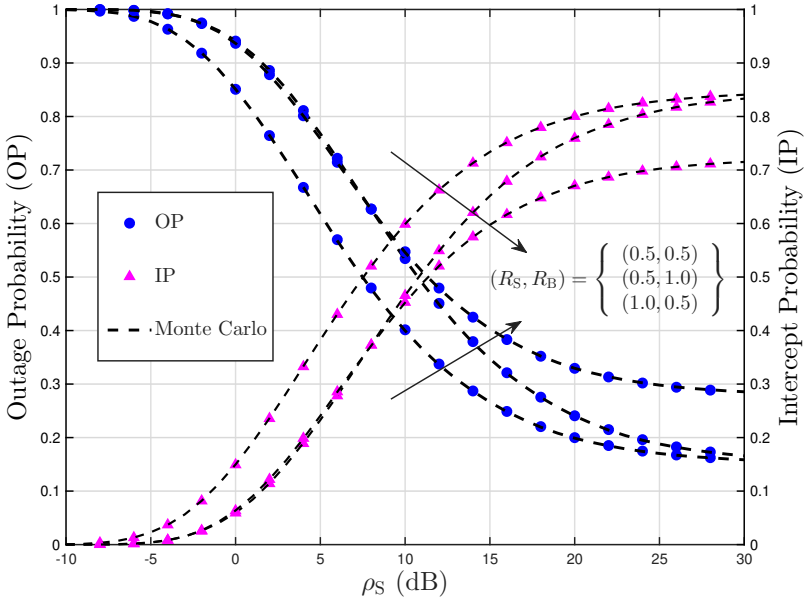


Figure 9. Reliability and security tradeoff in terms of OP and IP versus the normalized source transmit power ρ_S for different source and backscatter target-rate pairs $(R_S, R_B) \in \{(0.5, 0.5), (0.5, 1.0), (1.0, 0.5)\}$ bit/s/Hz, with $\rho_E = 5$ dB.

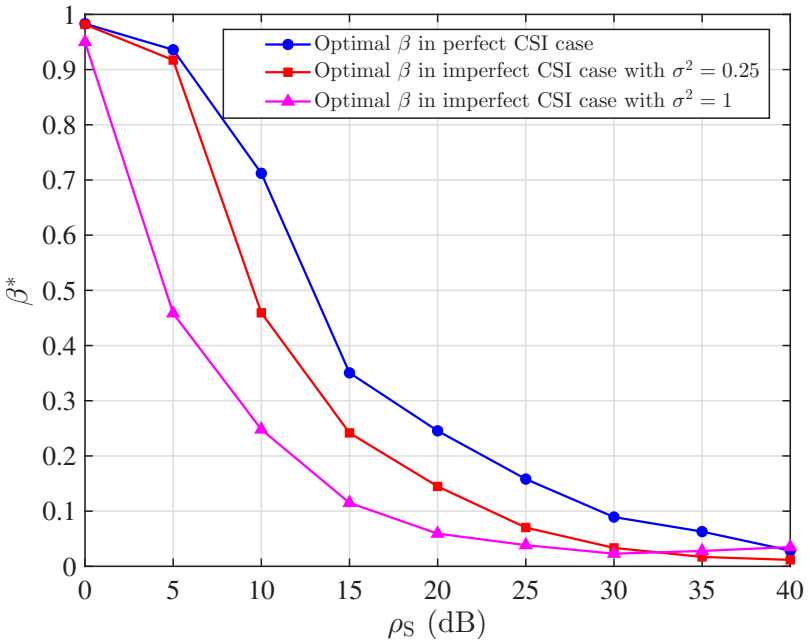


Figure 10. Reliability-oriented optimal backscatter coefficient $\beta^{\star}_{OP}$ versus ρ_S under perfect and imperfect CSI, with $\rho_E = 5$ dB, and $\sigma^2 \in \{0, 0.25, 1\}$.

Fig. 10 shows the reliability-oriented optimal backscatter coefficient $\beta^{\star}_{OP}$ versus ρ_S under perfect and imperfect CSI. In practice, perfect CSI is difficult to obtain because of channel estimation errors. Following the CSI estimation treatment adopted in[50], the estimated channel is modeled as $\widehat{h}_i = h_i + e_i$, where $\widehat{h}_i$ denotes the estimated channel coefficient, h_i is the corresponding actual channel coefficient, and e_i represents a zero-mean complex Gaussian channel estimation

error with variance σ^2. A common normalized error variance σ^2 is considered for the involved links, with $\sigma^2 = 0$ corresponding to perfect CSI. For the decoding process at D, $i \in \{\mathrm{SD}, \mathrm{SB}, \mathrm{BD}, \mathrm{ED}\}$. Consequently, the instantaneous SINRs under imperfect CSI are $\mathrm{SINR}_{\mathrm{D},1}^{\mathrm{imCSI}} = \rho_{\mathrm{S}} \left(\gamma_{\mathrm{SD}} + \sigma^2\right) / \left(\beta \rho_{\mathrm{S}} \left(\gamma_{\mathrm{SB}} + \sigma^2\right) \left(\gamma_{\mathrm{BD}} + \sigma^2\right) + \rho_{\mathrm{E}} \left(\gamma_{\mathrm{ED}} + \sigma^2\right) + 1\right)$ and $\mathrm{SINR}_{\mathrm{D},2}^{\mathrm{imCSI}} = \beta \rho_{\mathrm{S}} \left(\gamma_{\mathrm{SB}} + \sigma^2\right) \left(\gamma_{\mathrm{BD}} + \sigma^2\right) / \left(\rho_{\mathrm{E}} \left(\gamma_{\mathrm{ED}} + \sigma^2\right) + 1\right)$. Based on these SINRs, the same numerical search is performed to determine the value of β that minimizes the OP for each ρ_{S}. Under perfect CSI, $\beta_{\mathrm{OP}}^{\star}$ decreases as ρ_{S} increases. At low ρ_{S}, a relatively large reflection coefficient is required to provide sufficient received power for decoding the backscatter component in the second SIC stage. As ρ_{S} increases, this decoding requirement can be satisfied with a smaller β. Reducing β also decreases the backscatter-induced interference encountered during first-stage source decoding. Consequently, the reliability-oriented operating point progressively shifts toward smaller reflection coefficients. Under imperfect CSI, increasing σ^2 further shifts $\beta_{\mathrm{OP}}^{\star}$ toward the lower-β region, particularly at low and moderate ρ_{S}. This behavior can be interpreted from the above SINRs because $\left(\gamma_{\mathrm{SB}} + \sigma^2\right) \left(\gamma_{\mathrm{BD}} + \sigma^2\right)$ simultaneously determines the useful backscatter power in the second SIC stage and the backscatter-induced interference in the first SIC stage. Channel estimation errors therefore modify the balance between these two decoding requirements and consequently change the reliability-oriented operating point.

Fig. 11 depicts the interception-critical backscatter coefficient $\beta_{\mathrm{IP}}^{\star}$, defined as the value of β that maximizes the IP, versus ρ_{S} under perfect and imperfect CSI. Following the same CSI estimation treatment as in Fig. 10, the instantaneous SINRs under imperfect CSI are $\mathrm{SINR}_{\mathrm{E},1}^{\mathrm{imCSI}} = \rho_{\mathrm{S}} \left(\gamma_{\mathrm{SE}} + \sigma^2\right) / \left(\beta \rho_{\mathrm{S}} \left(\gamma_{\mathrm{SB}} + \sigma^2\right) \left(\gamma_{\mathrm{BE}} + \sigma^2\right) + \rho_{\mathrm{E}} \left(\gamma_{\mathrm{EE}} + \sigma^2\right) + 1\right)$ and $\mathrm{SINR}_{\mathrm{E},2}^{\mathrm{imCSI}} = \beta \rho_{\mathrm{S}} \left(\gamma_{\mathrm{SB}} + \sigma^2\right) \left(\gamma_{\mathrm{BE}} + \sigma^2\right) / \left(\rho_{\mathrm{E}} \left(\gamma_{\mathrm{EE}} + \sigma^2\right) + 1\right)$. Based on these SINRs, the numerical search is performed to determine the value of β that maximizes the IP for each ρ_{S}. Similar to the reliability-oriented result in Fig. 10, $\beta_{\mathrm{IP}}^{\star}$ generally decreases as ρ_{S} increases. At low ρ_{S}, successful interception is strongly constrained by second-stage backscatter decoding, which requires a relatively large reflection coefficient. As ρ_{S} increases, the backscatter component can be decoded with a smaller β. A smaller β simultaneously reduces the backscatter-induced interference during first-stage source decoding. The imperfect-CSI results further show that increasing σ^2 generally shifts $\beta_{\mathrm{IP}}^{\star}$ toward smaller values, particularly in the low and moderate ρ_{S} regions. This behavior follows from the dual role of $\left(\gamma_{\mathrm{SB}} + \sigma^2\right) \left(\gamma_{\mathrm{BE}} + \sigma^2\right)$ in the above SINRs because it contributes to the useful backscatter signal in the second SIC stage while also appearing as interference in the first SIC stage. CSI estimation errors therefore change the balance between the two decoding stages and consequently alter the backscatter coefficient at which the eavesdropper attains its most favorable interception condition.

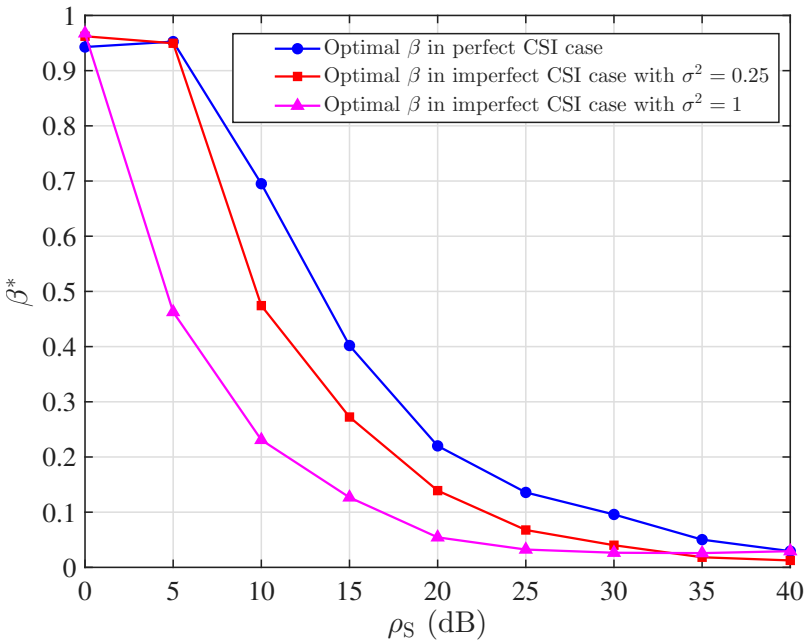


Figure 11. IP-maximizing backscatter coefficient $\beta_{\mathrm{IP}}^{\star}$ versus ρ_{S} under perfect and imperfect CSI, with $\rho_{\mathrm{E}} = 5$ dB, and $\sigma^2 \in \{0, 0.25, 1\}$.

6 Conclusion

This paper investigated the reliability and security performance of a secure source-assisted BC system in the presence of an active FD eavesdropper. Unlike passive eavesdropping models, the considered eavesdropper simultaneously jams the legitimate

destination and intercepts confidential information while suffering from residual self-interference. By accounting for two-stage SIC at both the legitimate destination and the eavesdropper, we derived the SINR expressions for the source and backscatter signals. Based on these expressions, exact analytical formulations for the OP and IP were obtained and further transformed into closed-form representations involving the bivariate generalized Fox-H function. High-SNR asymptotic expressions were also derived to characterize the limiting system behavior.

The analytical and Monte Carlo simulation results confirmed the accuracy of the proposed performance analysis. The results showed that increasing the source transmit power improves reliability by reducing the OP, but it also strengthens the eavesdropper's received signal and hence increases the IP. Moreover, both OP and IP converge to non-zero asymptotic limits in the high-SNR regime, indicating that transmit-power enhancement alone cannot remove the effects of interference-limited SIC operation. The backscatter coefficient was shown to have a non-monotonic impact on both reliability and security, since a larger reflection coefficient enhances the backscatter link but also increases co-channel interference during the first SIC stage. In addition, the eavesdropper's jamming power was found to play a dual role: it degrades the legitimate link and increases the OP, while simultaneously reducing the eavesdropper's own interception capability due to residual self-interference. These findings highlight the importance of jointly accounting for the source transmit power, backscatter coefficient, target data rates, and expected jamming level when designing secure source-assisted BC systems.

Future work will extend the present perfect-SIC analytical framework to practical imperfect SIC by explicitly accounting for residual post-cancellation interference in the joint decoding process and deriving the corresponding exact OP and IP expressions, closed-form representations, and high-SNR asymptotic characterizations. Another important direction is to extend the present GSS-based scalar search toward a constrained optimization framework that jointly determines the source transmit power and backscatter coefficient under practical reliability and security requirements.

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Author contributions statement

Minh-Han H. Do and Tan N. Nguyen proposed the main ideas, developed the mathematical derivations, and performed the Monte Carlo simulations. Thu-Ha Thi Pham and Phuong T. Tran prepared the writing draft. Finally, Hye-Young Kim, Byung-Seo Kim, and Miroslav Voznak supervised the overall work, and double-checked the final version of the manuscript. All authors reviewed and approved the final manuscript.

Data Availability

No datasets were generated or analysed during the current study.

Competing Interests

The authors declare no competing interests.

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